

Further Identities and Critical Transitions in the Polylogarithm–Stieltjes Calculus

Translation geometry, Euler reversal, Lerch velocities,
and depth in the Cayley shuffle quotient

Research continuation prepared for Vladimir Reshetnikov
For integration into the PROVEIT project

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Abstract

We continue the polylogarithm–Stieltjes program at a pinned ProveIt repository snapshot. A holomorphic translation identity for Hurwitz zeta generates all-order pairings of balanced Stieltjes antiderivatives, explicit logarithmic endpoint polynomials, a positive Gram kernel, and sharp Sobolev thresholds. Its first spectral derivative yields three exact formulas for the periodic Gamma translation energy, strict concavity with a unique half-period maximum, and a geometrically convergent odd-zeta identity for the first Stieltjes constant. An explicit epsilon-sensitive density estimate proves the existing near-critical Euler reversal conjecture, with a first logarithmic correction, Lambert approximation, derivative-zero ratios, and a universal transition profile. For diagonal Lerch velocities we determine the least eventual sign-block length and all-order sharp-cutoff finite parts, including the essential conjugate-singularity terms. Finally, an oriented shuffle-algebra retraction for the full Cayley ideal preserves depth and gives exact filtered dimensions. The mixed S_6 and current S_8 period candidates remain conjectural. Proofs, explicit source-status decisions, reproducible algebra and numerical diagnostics, and a proposed research agenda accompany the article.

Keywords. Polylogarithm; Hurwitz zeta; Gamma function; polygamma; generalized Stieltjes constants; Euler transformation; Lambert inverse; shuffle algebra; critical asymptotics.

Contents

1	Scope, principal results, and relation to the source material	3
1.1	Results at a glance	3
1.2	Classical antecedents and what is claimed here	4
1.3	Conventions and logical boundaries	4
2	A translation calculus for Hurwitz jets	4
2.1	Normalization and analytic justification	4
2.2	The exact master identity	5
2.3	Convergent endpoint expansion at every order	6
2.4	A positive integral for the endpoint kernel	7
2.5	Higher antiderivatives and sharp regularity	9
2.6	Antiderivatives with respect to the translation	10
3	A sharp Gamma translation theorem and explicit zeta identities	10
3.1	Three exact representations	10
3.2	Strict concavity and the optimal translation	11
3.3	Explicit derivatives, polygamma forms, and a definite primitive	12
3.4	A convergent identity for the first Stieltjes constant	13
4	A uniform transition at outer order one	14
4.1	Definitions and the inherited sign theorem	14
4.2	The reversal asymptotic and its correction	15
4.3	Preserving the vanishing algebraic amplitude	16
4.4	A joint expansion under the Euler kernel	17
4.5	Locating and expanding the unique zero	19
4.6	A universal profile and the positive maximum	19
4.7	Verification, integration status, and further questions	21
5	Sharp sign blocks for diagonal Lerch velocities	21
6	Higher critically weighted sums and their finite constants	23
7	A Cayley normal form that preserves series depth	26
7.1	Alphabet, admissibility, and the two counts	26
7.2	A count-preserving endpoint retraction	27
7.3	Eulerian lifts and the oriented projector	28
7.4	The exact depth filtration of the full ideal quotient	29
7.5	Completeness of the unmultiplied balanced rows	31
7.6	Consequences for the frozen S_6 search and verification	32
8	Source audit, proposed corrections, and integration decisions	33
8.1	Concrete source actions	33
8.2	Proof obligations that should remain explicit	33
8.3	Suggested placement	34
9	Further research questions and proposed work	34
9.1	Translation identities and the Stieltjes hierarchy	34
9.2	Uniform Euler limits	35
9.3	Lerch phase arithmetic and critical sums	36
9.4	Exact polylogarithmic reductions and formalization	36
10	Reproducibility and verification record	37

1 Scope, principal results, and relation to the source material

This report continues the polylogarithm manuscript and the active incoming research of the PROVEIT project. It is based on the fixed repository commit

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The consolidated manuscript and the newer continuation reports must be read together: several questions still visible in older fragments have already been settled in later reports. The source comparison, exact paths, and proposed integration actions are supplied in the accompanying claim ledger and provenance manifest.

The results here fall into four connected developments. The first is a translation calculus for the Hurwitz order jets that generate balanced Stieltjes antiderivatives. The second resolves an explicit joint-limit conjecture for Gaussian polylogarithmic Euler remainders. The third studies the exact signs and critical sums of the Lerch deformation velocities already identified in the latest continuation. The fourth gives an algebraic reduction preserving depth in the formal Cayley quotient. Each development has its own hypotheses; none is used as a substitute for an unproved arithmetic relation among numerical periods.

1.1 Results at a glance

1. **Exact translated Hurwitz pairings.** Theorem 2.1 expresses the bilinear periodic translation of two Hurwitz functions as a Gamma–trigonometric kernel times a symmetric Hurwitz difference. Its polylogarithmic form permits every order derivative. Theorem 2.3 identifies the complete logarithmic endpoint polynomial and an ordinary convergent even power series remainder at every derivative order.
2. **A sharp Gamma translation inequality.** The squared L^2 change in $\log \Gamma$ under a periodic translation is strictly concave on $(0, 1)$ and has its unique maximum at $1/2$ (Theorem 3.2). Its three exact representations yield an accelerated odd-zeta series for γ_1 with an explicit geometric tail (Corollary 3.3). Higher derivatives connect this energy to polygamma functions and Hurwitz order derivatives; explicit primitives follow from the same kernel.
3. **Proof of the near-critical Euler conjecture.** For outer order $a = 1 - \varepsilon$, fixed inner order $b > 0$, and $\lambda = \log(1/\varepsilon)$, the unique reversal index satisfies

$$\nu(1 - \varepsilon, b) \sim \frac{\lambda^{2b+2}}{\pi b^2 \Gamma(b+1)^2 \zeta(b+1)^2 \varepsilon^2}.$$

Theorem 4.1 proves the source conjecture `research:conj:crossing`, including a first inverse-logarithmic correction uniformly for b in compact positive intervals. The proof retains the exponentially small seed and an ε factor in the algebraic remainder. A rescaled profile is $x^{-1/2} - x^{-1}$; the later maximum occurs at an index asymptotic to 4ν .

4. **Optimal sign blocks and all-order critical finite parts.** For the Lerch velocities $v_n(A)$, let $\theta(A)$ be the phase of the dominant inverse singularity. The least eventual block length containing both signs is exactly $\lceil \pi/\theta \rceil + 1$ (Theorem 5.1). At $A = 2$ it is four. Theorem 6.1 computes the finite part of every sharp sum $\sum_{n \leq N} n^p v_n(A) \tau^n$. Both conjugate singularities are required for the divergent subtractions once $p \geq 2$.
5. **A depth-preserving formal Cayley reduction.** An oriented choice of shuffle generators yields a multiplicative retraction whose kernel is the full Cayley ideal and whose output does not increase depth. Its generator counts give the exact depth filtration of the quotient. This is a structural algebraic result; the mixed S_6 period identity remains conjectural.

1.2 Classical antecedents and what is claimed here

Hurwitz’s Fourier expansion, the parameter recurrence, and the order-zero Gamma evaluation are classical [1]. Espinosa and Moll developed product integrals and antiderivatives involving Hurwitz zeta [3, 4], and a generalized polygamma framework [5]. These works supply the appropriate context for the first two sections. We give a direct analytic derivation of the translated identities and their sharp consequences; the underlying unshifted product integral is explicitly credited. The standard polylogarithm conventions are those summarized in [2].

The Euler section uses the exact density and the global one-crossing theorem already proved in the newest applicable source [11]. Its new work is the uniform estimate in the singular joint limit, the corrected root law, and the profile consequences. The Lerch sections use the selected-sheet continuation and additive coefficient asymptotic from [12]; they answer two further questions posed there. Lagrange inversion and singularity transfer are standard tools [7, 6]. Earlier Lambert polynomial and branch expansions provide further context [8, 9].

Throughout, “new” means established here relative to the checked repository baseline. This source audit is not a claim of worldwide priority for every identity or special case. The classical tools are identified, inherited repository theorems are marked, and new assertions are supplied with proofs. The unresolved S_6 and current S_8 candidates retain their conjectural status even when numerical residuals are tiny.

1.3 Conventions and logical boundaries

The Laurent convention is $\zeta(1+u, a) = u^{-1} + \sum_{n \geq 0} (-1)^n \gamma_n(a) u^n / n!$; in particular $\gamma_0(a) = -\psi(a)$. Superscripts on ζ denote derivatives with respect to its order, while superscripts on γ_n and ψ denote derivatives with respect to their real argument. Rising factorials are $(z)_m$ and falling factorials are $(z)_{\underline{m}}$. Endpoint values of periodic L^2 functions are understood almost everywhere. Branches of complex powers and square roots are fixed where first used.

The analytic arguments, exact rational algebra, and floating-point diagnostics play different roles. Proofs justify every limiting operation that enters a theorem. Exact finite calculations check implementations and stated ranks in specified spaces. Floating-point calculations test formulas independently and illustrate asymptotic accuracy; they are not interval certificates. The package does not claim proof-assistant formalization or arithmetic independence from numerical experiments.

2 A translation calculus for Hurwitz jets

The parameter identity $\partial_x \zeta(s, x) = -s \zeta(s+1, x)$ connects the three central objects of this section: Hurwitz order derivatives, Stieltjes antiderivatives, and the logarithm of the Gamma function. We add translation in x as a second operation. The resulting bilinear identity has both a polylogarithmic form and a Hurwitz form. The latter makes all endpoint logarithms and all derivative orders explicit.

2.1 Normalization and analytic justification

Write

$$\zeta(1+u, x) = \frac{1}{u} + \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \gamma_n(x) u^n, \quad x > 0. \quad (2.1)$$

Order derivatives of ζ always refer to its first argument. Define

$$F_r(x) = \zeta^{(r)}(0, x), \quad J_n(x) = \frac{(-1)^{n+1}}{n+1} F_{n+1}(x) \quad (r, n \geq 0). \quad (2.2)$$

Then

$$F_0(x) = \frac{1}{2} - x, \quad F_1(x) = \log \Gamma(x) - \frac{1}{2} \log(2\pi), \quad J'_n(x) = \gamma_n(x).$$

The last equality follows by comparing coefficients in $\partial_x \zeta(s, x) = -s\zeta(s+1, x)$ at $s = 0$. This classical ladder is already present in the manuscript; its earlier literature is discussed there. Our normalization is the mean-zero one:

$$\int_0^1 F_r(x) dx = \int_0^1 J_n(x) dx = 0. \quad (2.3)$$

Thus J_n differs by a specified constant from the manuscript's primitive normalized to vanish at 1.

All functions in (2.2) are regarded as periodic functions almost everywhere on $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. Endpoint values themselves are immaterial. Put

$$\Delta_h f(x) = f(\{x+h\}) - f(x), \quad 0 < h < 1.$$

This is a periodic translation, not a translation of the original nonperiodic Gamma function on the whole positive half-line.

For completeness, the analytic justification for all forthcoming coefficient operations can be made at the level of Banach-space-valued holomorphic functions. On every disk $|s| \leq \rho < 1/2$,

$$\zeta(s, x) = x^{-s} + \zeta(s, 1+x), \quad 0 < x < 1,$$

and the second term and its order derivatives are uniformly bounded on smaller disks. Consequently the map $s \mapsto \zeta(s, \cdot)$ is holomorphic with values in $L^2(0, 1)$; its r th derivative is dominated by a constant times $1 + x^{-\rho} |\log x|^r$. Translation is a bounded operator on L^2 . The bilinear products below are therefore jointly holomorphic, and order differentiation under the integrals is valid. The identity $\int_0^1 \zeta(s, x) dx = 0$ for $\Re s < 1$ proves (2.3) by the same argument.

2.2 The exact master identity

For $u = s + t$ set

$$\mathcal{R}(s, t) = \frac{\Gamma(1-s)\Gamma(1-t)}{\Gamma(2-s-t)} \frac{\cos(\pi(s-t)/2)}{\cos(\pi(s+t)/2)}, \quad (2.4)$$

$$\mathcal{Z}(u, h) = \zeta(u-1, h) + \zeta(u-1, 1-h) - 2\zeta(u-1). \quad (2.5)$$

Both expressions are holomorphic near $(s, t) = (0, 0)$.

Theorem 2.1 (Exact translation identity). *For $0 < h < 1$ and $|s|, |t| < 1/4$,*

$$\boxed{\int_0^1 \Delta_h \zeta(s, x) \Delta_h \zeta(t, x) dx = \mathcal{R}(s, t) \mathcal{Z}(s+t, h).} \quad (2.6)$$

In particular, for every $r, q \geq 0$,

$$\int_0^1 \Delta_h F_r(x) \Delta_h F_q(x) dx = \partial_s^r \partial_t^q (\mathcal{R}(s, t) \mathcal{Z}(s+t, h)) \big|_{s=t=0}. \quad (2.7)$$

Multiplication by $(-1)^{m+n+2}/((m+1)(n+1))$ gives the corresponding identity for J_m, J_n when $r = m+1, q = n+1$.

Proof. Use the Fourier convention $\widehat{f}(k) = \int_0^1 f(x) e^{-2\pi i k x} dx$. Hurwitz's Fourier formula gives

$$\widehat{\zeta(s, \cdot)}(0) = 0, \quad \widehat{\zeta(s, \cdot)}(k) = \Gamma(1-s)(2\pi i k)^{s-1} \quad (k \neq 0), \quad (2.8)$$

where $\log(2\pi ik) = \log(2\pi|k|) + \frac{\pi i}{2} \operatorname{sgn} k$. Initially one may take $\Re s < 0$ and then extend in L^2 . These are classical Fourier coefficients [1, 3]. Parseval's identity for the bilinear pairing, together with Δ_h acting by $e^{2\pi i k h} - 1$, yields

$$\begin{aligned} \mathcal{D}(s, t; h) &= 4\Gamma(1-s)\Gamma(1-t)(2\pi)^{u-2} \cos \frac{\pi(s-t)}{2} \\ &\quad \cdot \left\{ \zeta(2-u) - \frac{\operatorname{Li}_{2-u}(e^{2\pi i h}) + \operatorname{Li}_{2-u}(e^{-2\pi i h})}{2} \right\}. \end{aligned} \quad (2.9)$$

The series on this line are absolutely and locally uniformly convergent in the stated bidisk. Notice that the symmetric pair of polylogarithms must be retained when s, t are complex; writing a real part would then destroy holomorphic dependence on the orders.

The even part of the same Hurwitz Fourier formula, now at order $u-1$, is

$$\begin{aligned} \zeta(u-1, h) + \zeta(u-1, 1-h) &= -4\Gamma(2-u)(2\pi)^{u-2} \cos \frac{\pi u}{2} \frac{\operatorname{Li}_{2-u}(e^{2\pi i h}) + \operatorname{Li}_{2-u}(e^{-2\pi i h})}{2}, \\ 2\zeta(u-1) &= -4\Gamma(2-u)(2\pi)^{u-2} \cos \frac{\pi u}{2} \zeta(2-u). \end{aligned}$$

Their difference converts (2.9) into (2.6). The L^2 argument above proves the differentiated identity. $\square$

Remark 2.2 (Relation to the classical product integral). At zero shift the underlying correlation is the classical identity

$$\int_0^1 \zeta(s, x) \zeta(t, x) dx = -\mathcal{R}(s, t) \zeta(s+t-1),$$

given, for example, by Espinosa–Moll [3, Theorem 3.1]. We do not claim this antecedent as new. The contribution used here is the translated difference, its uniformly justified jet calculus, and the explicit consequences below. No claim of global priority is made for every special-case identity obtained by differentiating the master formula.

2.3 Convergent endpoint expansion at every order

Put $\ell = \log(1/h)$. The recurrence for Hurwitz zeta gives

$$\mathcal{Z}(u, h) = h^{1-u} + \zeta(u-1, 1+h) + \zeta(u-1, 1-h) - 2\zeta(u-1). \quad (2.10)$$

The last three terms are analytic and even in h for $|h| < 1$. Expanding them around $h = 0$ gives the particularly useful formula

$$\begin{aligned} \mathcal{D}(s, t; h) &= \mathcal{R}(s, t) \left\{ h^{1-u} + u(u-1)\zeta(1+u)h^2 \right. \\ &\quad \left. + 2 \sum_{j=2}^{\infty} \frac{(u-1)_{2j}}{(2j)!} \zeta(2j-1+u)h^{2j} \right\}. \end{aligned} \quad (2.11)$$

The product $u(u-1)\zeta(1+u)$ has its removable value -1 at zero. This cancellation is performed before differentiation.

Theorem 2.3 (All-order logarithmic endpoint polynomial). *For fixed $r, q \geq 0$ there is a monic polynomial of degree $r+q$,*

$$P_{r,q}(\ell) = \partial_s^r \partial_t^q (\mathcal{R}(s, t) e^{(s+t)\ell}) \Big|_{s=t=0}, \quad (2.12)$$

such that

$$\int_0^1 \Delta_h F_r \Delta_h F_q dx = h P_{r,q}(\log(1/h)) + \sum_{j=1}^{\infty} a_{r,q,j} h^{2j}, \quad 0 < h < 1. \quad (2.13)$$

The even series is convergent, locally uniformly for $|h| < 1$. If $r + q \geq 1$, the coefficient of ℓ^{r+q-1} is $r + q$. Every coefficient of $P_{r,q}$ lies in the algebra over $\mathbb{Q}$ generated by $\zeta(2), \dots, \zeta(r + q)$; Euler's constant cancels. The coefficient $a_{r,q,1}$ uses only Stieltjes constants through γ_{r+q-1} and these zeta values. For $j \geq 2$, it uses derivatives of ζ at $2j - 1$ of order at most $r + q - 1$. For $r = q = 0$ the statements about negative derivative indices mean that only the constant $a_{0,0,1} = -1$ is present.

Proof. Equation (2.11), differentiated in the bidisk, gives (2.13). Uniform convergence with order derivatives follows also directly from the analytic Taylor expansion in (2.10) on any smaller circle $|h| \leq h_0 < 1$. The required coefficients of $\mathcal{R}$ are especially transparent in its logarithm:

$$\begin{aligned} \log \mathcal{R}(s, t) &= \sum_{k \geq 1} \frac{(s+t)^k}{k} + \sum_{k \geq 2} \frac{\zeta(k)}{k} (s^k + t^k - (s+t)^k) \\ &\quad + \sum_{m \geq 1} \frac{(1 - 2^{-2m})\zeta(2m)}{m} ((s+t)^{2m} - (s-t)^{2m}). \end{aligned} \quad (2.14)$$

This follows from $\Gamma(2-u) = (1-u)\Gamma(1-u)$ and the convergent logarithmic Gamma and cosine products. In particular, $\mathcal{R}(0,0) = 1$ and $\partial_s \mathcal{R}(0,0) = \partial_t \mathcal{R}(0,0) = 1$. These facts give the two leading coefficients of P . The same formula proves the claimed coefficient algebra. Finally $(u-1)_{2j}$ has a simple zero at $u = 0$ for every $j \geq 1$. At $j = 1$ it removes the pole of $\zeta(1+u)$; at $j \geq 2$ it lowers the highest needed zeta derivative by one. $\square$

Here are the first nontrivial endpoint polynomials:

$$P_{1,1}(\ell) = \ell^2 + 2\ell + 2 + \frac{\pi^2}{3}, \quad (2.15)$$

$$P_{1,2}(\ell) = \ell^3 + 3\ell^2 + \left(6 + \frac{2\pi^2}{3}\right)\ell + 6 + \frac{2\pi^2}{3} - 2\zeta(3), \quad (2.16)$$

$$\begin{aligned} P_{2,2}(\ell) &= \ell^4 + 4\ell^3 + \left(12 + \frac{4\pi^2}{3}\right)\ell^2 + \left(24 + \frac{8\pi^2}{3} - 8\zeta(3)\right)\ell \\ &\quad + 24 + \frac{8\pi^2}{3} - 8\zeta(3) + \frac{7\pi^4}{45}. \end{aligned} \quad (2.17)$$

The supplied exact-arithmetic program constructs these polynomials and additional ones from (2.14). Formula (2.13) is stronger than a finite asymptotic expansion: after the single logarithmic term, the remaining function is an ordinary convergent even power series.

2.4 A positive integral for the endpoint kernel

The Gamma–trigonometric factor has an elementary positive-kernel interpretation, which gives information about all the endpoint polynomials simultaneously.

Theorem 2.4 (Endpoint Gram identity). *For $|s|, |t| < 1/4$, with $u = s + t$,*

$$\mathcal{R}(s, t) = \frac{1}{1-u} + \int_0^\infty [(x+1)^{-s} - x^{-s}][(x+1)^{-t} - x^{-t}] dx. \quad (2.18)$$

For every real ℓ this implies

$$\begin{aligned} P_{r,q}(\ell) &= \int_0^1 (\ell - \log x)^{r+q} dx \\ &\quad + \int_0^\infty [(\ell - \log(x+1))^r - (\ell - \log x)^r] \\ &\quad \cdot [(\ell - \log(x+1))^q - (\ell - \log x)^q] dx. \end{aligned} \quad (2.19)$$

Every finite principal matrix of $(P_{r,q}(\ell))_{r,q \geq 0}$ is positive definite. For the initial blocks $P_N(\ell) = (P_{r,q}(\ell))_{0 \leq r,q \leq N}$,

$$\det P_N(\ell) = \det P_N(0) \geq \prod_{r=0}^N (r!)^2, \quad (2.20)$$

with strict inequality for $N \geq 1$.

Proof. The integral in (2.18) is locally holomorphic: at infinity its integrand is $O(x^{-\Re u-2})$, with fixed logarithmic factors after differentiation, and at zero the factors are bounded by a constant times $1 + x^{-\rho}$ for any fixed $\rho < 1/2$ on a smaller bidisk. Make the substitution $y = x/(1+x)$. The integral becomes

$$\int_0^1 (1-y)^{u-2} [1 - y^{-s} - y^{-t} + y^{-u}] dy.$$

The bracket has a double zero at $y = 1$. To evaluate it without separating divergent integrals, use the twice-subtracted beta formula

$$B(a, b) = \int_0^1 (1-y)^{b-1} [y^{a-1} - 1 + (a-1)(1-y)] dy \\ + \frac{1}{b} - \frac{a-1}{b+1}.$$

It follows from the ordinary beta integral when $\Re b > 0$ and continues to $\Re a > 0$, $\Re b > -2$, away from its displayed poles. Apply it to $a = 1, 1-s, 1-t, 1-u$ with signs $+, -, -, +$ and $b = u-1$. Both subtraction terms cancel. Thus the original integral equals

$$-\frac{1}{1-u} - \Gamma(u-1) \left[\frac{\Gamma(1-s)}{\Gamma(t)} + \frac{\Gamma(1-t)}{\Gamma(s)} \right].$$

The term $B(1-u, u-1)$ is zero under this meromorphic continuation. Reflection of the Gamma factors, followed by

$$\frac{\sin \pi s + \sin \pi t}{\sin \pi(s+t)} = \frac{\cos(\pi(s-t)/2)}{\cos(\pi(s+t)/2)},$$

identifies the last expression with $\mathcal{R}(s, t) - (1-u)^{-1}$. All apparent singularities at $u = 0$ are removable in the combined identity, proving (2.18) throughout the bidisk.

Multiply by $e^{u\ell}$ and differentiate at zero to obtain (2.19). The two terms are Gram matrices of real functions. The first is strictly positive definite for any distinct finite set of indices: a nonzero polynomial in $\ell - \log x$ cannot vanish almost everywhere on $(0, 1)$.

Let $T(\ell)_{r,j} = \binom{r}{j} \ell^{r-j}$ for $j \leq r$ and zero otherwise. The binomial theorem gives the congruence $P_N(\ell) = T(\ell)P_N(0)T(\ell)^T$ and $\det T(\ell) = 1$. For the first Gram term alone the matrix at $\ell = 0$ is $((r+q)!)$. Its determinant is $\prod_{r=0}^N (r!)^2$, because

$$\binom{r+q}{r} = \sum_k \binom{r}{k} \binom{q}{k}$$

is a factorization by a unit lower triangular Pascal matrix and its transpose. Adding a positive semidefinite matrix cannot decrease the determinant of a positive definite matrix. The second term of (2.19) is nonzero for $N \geq 1$, as its $(1, 1)$ entry is the integral of $\log^2(1+1/x) > 0$. Hence the determinant increases strictly. This proves (2.20). The shift-invariance assertion concerns the initial blocks; arbitrary principal submatrices need not have shift-invariant determinants. $\square$

For example, $\det P_1(\ell) = 1 + \pi^2/3$. The 2×2 case of the Gram positivity gives $P_{r,q}(\ell)^2 < P_{r,r}(\ell)P_{q,q}(\ell)$ for distinct r, q . These inequalities hold for every real ℓ , not merely for the large positive logarithms arising in a small shift.

2.5 Higher antiderivatives and sharp regularity

For $m \geq 0$ define

$$F_{r,m}(x) = \partial_s^r \frac{\zeta(s-m, x)}{(1-s)_m} \Big|_{s=0}, \quad (1-s)_0 = 1. \quad (2.21)$$

The identity $\partial_x \zeta(s-m, x) = (m-s)\zeta(s-m+1, x)$ shows $\partial_x F_{r,m} = F_{r,m-1}$ for $m \geq 1$. These are the uniquely normalized successive periodic primitives: every one has mean zero. In particular

$$\frac{(-1)^{n+1}}{n+1} F_{n+1,m-1} \quad (m \geq 1)$$

is an m -fold antiderivative of γ_n on $(0, 1)$. The mean-zero convention removes all polynomial ambiguities.

Theorem 2.5 (Exact Sobolev thresholds). *For each $r, m \geq 0$ and real σ ,*

$$F_{r,m} \in H^\sigma(\mathbb{T}) \iff \sigma < m + \frac{1}{2}.$$

Thus the m -fold antiderivative of γ_n , with $m \geq 1$, has threshold $m - 1/2$. Moreover

$$\|\Delta_h J_n\|_2^2 \sim \frac{h \log^{2n+2}(1/h)}{(n+1)^2} \quad (h \downarrow 0). \quad (2.22)$$

For any nonzero finite combination $G = \sum_{n=0}^N c_n J_n$ with $c_N \neq 0$, the right side becomes $c_N^2 h \log^{2N+2}(1/h)/(N+1)^2$ when the coefficients are real.

Proof. Let

$$Q_r(\lambda) = \partial_s^r (\Gamma(1-s)e^{\lambda s}) \Big|_{s=0}.$$

This is a monic polynomial in λ of degree r ; for example $Q_1(\lambda) = \lambda + \gamma$ and $Q_2(\lambda) = (\lambda + \gamma)^2 + \zeta(2)$. Using $\Gamma(m+1-s)/(1-s)_m = \Gamma(1-s)$ in (2.8),

$$\widehat{F_{r,m}}(k) = \frac{Q_r(\log(2\pi i k))}{(2\pi i k)^{m+1}} \quad (k \neq 0). \quad (2.23)$$

The squared H^σ norm is therefore governed, up to positive constant factors for large k , by $\sum_{k \geq 2} k^{2\sigma-2m-2} \log^{2r} k$. The integral test proves precisely the stated threshold, including divergence at equality. Equation (2.22) follows from the monic polynomial in Theorem 2.3; in a finite combination its highest logarithmic degree cannot cancel. $\square$

Every finite Gram matrix

$$\left(\int_0^1 F_r(x) F_q(x) dx \right)_{0 \leq r, q \leq N}$$

is also strictly positive definite. If a linear combination vanished in L^2 , its Fourier coefficients would make a polynomial in $\log(2\pi k) + i\pi/2$ vanish for all positive integers k . That polynomial is identically zero; triangularity of the monic Q_r then forces every coefficient of the combination to vanish. This is functional linear independence in L^2 , not an assertion of arithmetic independence of special values.

2.6 Antiderivatives with respect to the translation

There is a second antiderivative operation: integrate the translated pairing in its shift, rather than integrate each factor in its argument. It also remains inside the Hurwitz hierarchy. For $m \geq 1$ put

$$\mathcal{A}_m(s, t; h) = \mathcal{R}(s, t) \left\{ \frac{\zeta(u-1-m, h) + (-1)^m \zeta(u-1-m, 1-h)}{(2-u)_m} - \frac{2\zeta(u-1)h^m}{m!} \right\}, \quad u = s + t. \quad (2.24)$$

Corollary 2.6 (The complete shift-antiderivative ladder). *For $|s|, |t| < 1/4$ and $0 < h < 1$, $\partial_h^m \mathcal{A}_m(s, t; h) = \mathcal{D}(s, t; h)$. The repeated definite integral is*

$$\begin{aligned} \frac{1}{(m-1)!} \int_0^h (h-v)^{m-1} \mathcal{D}(s, t; v) dv &= \mathcal{A}_m(s, t; h) \\ &- 2\mathcal{R}(s, t) \sum_{k=1}^{\lfloor m/2 \rfloor} \frac{\zeta(u-1-2k)}{(2-u)_{2k}} \frac{h^{m-2k}}{(m-2k)!}. \end{aligned} \quad (2.25)$$

Every finite derivative in s, t may be taken in this identity.

Proof. The parameter recurrence gives $\partial_h^m \zeta(u-1-m, h) = (2-u)_m \zeta(u-1, h)$. The reflected term acquires the additional factor $(-1)^m$, which cancels its displayed sign. Differentiating the polynomial term gives $-2\zeta(u-1)$, proving the indefinite identity. To normalize, use the Hurwitz shift formula as $h \downarrow 0$. The term h^{m+1-u} and its first $m-1$ derivatives vanish, and the remaining values at zero are

$$\partial_h^j \mathcal{A}_m(s, t; 0) = \mathcal{R}(s, t) \frac{[1 + (-1)^{m-j}] \zeta(u-1-m+j)}{(2-u)_{m-j}}, \quad 0 \leq j < m.$$

Subtracting their Taylor polynomial gives (2.25). The same endpoint bound with fixed powers of $\log h$ justifies all order derivatives. In particular the sum is empty for $m = 1$. $\square$

3 A sharp Gamma translation theorem and explicit zeta identities

The first spectral derivative already gives a concrete identity tying together an ordinary quadratic Gamma integral, polylogarithm order derivatives, negative Hurwitz jets, and the first Stieltjes constant. Define the periodic translation energy

$$V(h) = \int_0^1 [\log \Gamma(\{x+h\}) - \log \Gamma(x)]^2 dx, \quad 0 < h < 1. \quad (3.1)$$

The singularities are logarithmic and the integral converges ordinarily. We extend it continuously by $V(0) = V(1) = 0$.

3.1 Three exact representations

Put $Z_j(h) = \zeta^{(j)}(-1, h) + \zeta^{(j)}(-1, 1-h) - 2\zeta^{(j)}(-1)$. Since $Z_0(h) = h(1-h)$, differentiation of Theorem 2.1 yields the following explicit formula.

Theorem 3.1 (Gamma–Hurwitz–polylogarithm identity). *For $0 < h < 1$,*

$$V(h) = Z_2(h) + 2Z_1(h) + \left(2 + \frac{\pi^2}{3}\right) h(1-h). \quad (3.2)$$

Equivalently, set $c = \gamma + \log(2\pi)$ and

$$E_j(h) = \zeta^{(j)}(2) - \Re \partial_s^j \text{Li}_s(e^{2\pi i h}) \Big|_{s=2}.$$

Then

$$V(h) = \frac{(c^2 + \pi^2/4)E_0(h) - 2cE_1(h) + E_2(h)}{\pi^2}. \quad (3.3)$$

The following positive-odd-order series also converges for $0 < h < 1$:

$$\begin{aligned} V(h) = & h \left(\log^2 h - 2 \log h + 2 + \frac{\pi^2}{3} \right) + \left(2\gamma_1 - \frac{\pi^2}{3} \right) h^2 \\ & - \sum_{j=2}^{\infty} \frac{2[\zeta'(2j-1) + H_{2j-2}\zeta(2j-1)]}{j(2j-1)} h^{2j}. \end{aligned} \quad (3.4)$$

Here $H_m = \sum_{k=1}^m 1/k$ and $\gamma_1 = \gamma_1(1)$.

Proof. From (2.14), $\mathcal{R}_s(0,0) = \mathcal{R}_t(0,0) = 1$ and $\mathcal{R}_{st}(0,0) = 2 + \pi^2/3$. Thus (2.7) with $r = q = 1$ is exactly (3.2). Formula (3.3) follows either by differentiating (2.9) or directly from

$$\widehat{F}_1(k) = \frac{\gamma + \log(2\pi i k)}{2\pi i k}.$$

The order derivatives of Li_s are legitimate here because the corresponding series $\sum (\log k)^j/k^2$ converges absolutely.

For the series, use (2.11). At $j = 1$,

$$u(u-1)\zeta(1+u) = -1 + (1-\gamma)u + (\gamma + \gamma_1)u^2 + O(u^3).$$

The mixed derivative of this expression times $\mathcal{R}$ is $2\gamma_1 - \pi^2/3$. For $j \geq 2$,

$$(u-1)_{2j} = -(2j-2)! u[1 + (H_{2j-2} - 1)u + O(u^2)].$$

Multiplying by $\zeta(2j-1+u)$ and taking the mixed derivative after multiplication by $\mathcal{R}$ gives $-4[\zeta'(2j-1) + H_{2j-2}\zeta(2j-1)]/((2j)(2j-1))$. The first term is (2.15). This proves (3.4). $\square$

3.2 Strict concavity and the optimal translation

Theorem 3.2 (Sharp half-period maximum). *The function V is strictly concave on $(0,1)$ and symmetric about $1/2$. Its unique maximum is at $h = 1/2$, with exact value*

$$V_* = -3\zeta''(-1) + (2\log 2 - 6)\zeta'(-1) - \frac{\log^2 2 + 2\log 2}{12} + \frac{1}{2} + \frac{\pi^2}{12}. \quad (3.5)$$

In particular $V(h) \leq V_$ for every real periodic shift h , with equality exactly at $h \equiv 1/2 \pmod{1}$. Numerically, $V_* = 2.680722487980507384821524139517\dots$*

Proof. Twice differentiating (3.2), using the Laurent expansion at 1 before evaluating removable singularities, gives

$$V''(h) = 2 \left(\gamma_1(h) + \gamma_1(1-h) - \frac{\pi^2}{3} \right). \quad (3.6)$$

We prove a sufficient strict bound entirely analytically. The derivative tower gives, by an absolutely convergent series,

$$\gamma_1'(a) = \sum_{m=0}^{\infty} \frac{1 - \log(m+a)}{(m+a)^2}.$$

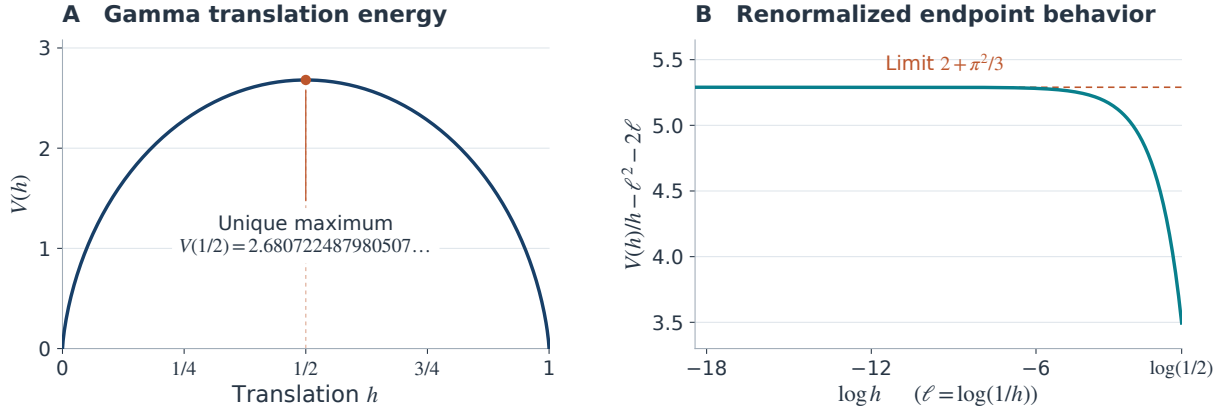


Figure 1: The Gamma translation energy and its renormalized endpoint limit. The left panel illustrates the proved strict concavity and half-period maximum. The right subtracts the explicit logarithmic terms, leaving a limit of $2 + \pi^2/3$. Curves are evaluated from the convergent series with its analytic truncation bound.

For $0 < a \leq 1$, its $m = 0$ term is at least 1, its $m = 1$ term is positive, and the possibly negative remaining terms have total magnitude at most $\sum_{m \geq 2} \log m/m^2$. The function $\log x/x^2$ is decreasing for $x \geq 2$, so

$$-\zeta'(2) \leq \frac{\log 2}{4} + \frac{\log 3}{9} + \int_3^\infty \frac{\log x}{x^2} dx = \frac{\log 2}{4} + \frac{4 \log 3}{9} + \frac{1}{3} < \frac{359}{360} < 1.$$

The strict rational bound uses $\log 2 < 7/10$ and $\log 3 < 11/10$. Hence $\gamma'_1(a) > 0$ on $(0, 1]$.

The sum-integral definition of γ_1 , and decrease of $\log x/x$ for $x \geq 3$, also give

$$\gamma_1(1) \leq \frac{\log 2}{2} + \frac{\log 3}{3} - \frac{\log^2 3}{2} < \frac{43}{60} < 1 < \frac{\pi^2}{6}.$$

Thus both Stieltjes constants in (3.6) are less than $\pi^2/6$, proving strict concavity. Periodic translation invariance gives $V(1-h) = V(h)$, so the unique maximum is at $1/2$. Finally insert $\zeta(s, 1/2) = (2^s - 1)\zeta(s)$ and its first two order derivatives into (3.2). This gives (3.5). The displayed decimal is a diagnostic evaluation; neither the sign proof nor optimality depends on it. $\square$

3.3 Explicit derivatives, polygamma forms, and a definite primitive

The exact Gamma energy also has useful identities beyond its second derivative. For every integer $m \geq 1$ and $0 < h < 1$,

$$V^{(m+2)}(h) = 2[\gamma_1^{(m)}(h) + (-1)^m \gamma_1^{(m)}(1-h)]. \quad (3.7)$$

The classical derivative tower takes the particularly simple form

$$\gamma_1^{(m)}(a) = (-1)^{m+1} m! \zeta'(m+1, a) + H_m \psi^{(m)}(a), \quad a > 0. \quad (3.8)$$

Indeed, compare the coefficient of u in

$$\partial_a^m \zeta(1+u, a) = (-1)^m (1+u)_m \zeta(m+1+u, a),$$

using $\partial_u(1+u)_m|_{u=0} = m!H_m$ and $\psi^{(m)}(a) = (-1)^{m+1} m! \zeta(m+1, a)$. Equations (3.7)–(3.8) therefore express all higher shift derivatives directly in polygamma functions and Hurwitz order

derivatives at positive integers. The identities apply on the open interval; the endpoint derivatives generally diverge.

For the primitive, define

$$W_j(h) = \zeta^{(j)}(-2, h) - \zeta^{(j)}(-2, 1-h), \quad a_0 = 2 + \frac{\pi^2}{3}.$$

Corollary 2.6, with $m = 1$ followed by $\partial_s \partial_t$ at zero, gives

$$\begin{aligned} \int_0^h V(v) dv &= \frac{W_2(h)}{2} + \frac{3W_1(h)}{2} + \left(\frac{7}{4} + \frac{\pi^2}{6} \right) W_0(h) \\ &\quad - 2h[\zeta''(-1) + 2\zeta'(-1) + a_0\zeta(-1)]. \end{aligned} \quad (3.9)$$

There is no omitted integration constant: all W_j vanish at zero by continuous extension, since $h^2 \log^j h \rightarrow 0$. At $h = 1$ this becomes the exact mean-energy identity

$$\int_0^1 V(h) dh = -2\zeta''(-1) - 4\zeta'(-1) + \frac{1}{3} + \frac{\pi^2}{18}. \quad (3.10)$$

It is also twice the centered second moment of $\log \Gamma$ on $(0, 1)$, as follows directly by averaging the defining translation.

3.4 A convergent identity for the first Stieltjes constant

The coefficients removed in (3.4) are all positive. Indeed, for $p \geq 3$,

$$H_{p-1}\zeta(p) + \zeta'(p) \geq \frac{3}{2} - \sum_{k \geq 2} \frac{\log k}{k^3} \geq \frac{3}{2} - \frac{\log 2}{4} - \frac{1}{16} > 0.$$

They are $O(\log j/j^2)$ after the displayed prefactor in (3.4). The series consequently extends uniformly to $0 \leq h \leq 1$. Letting $h \uparrow 1$ and using $V(1) = 0$ proves

$$\boxed{\gamma_1 = -1 + \sum_{j=2}^{\infty} \frac{\zeta'(2j-1) + H_{2j-2}\zeta(2j-1)}{j(2j-1)}}. \quad (3.11)$$

This form is slow but has positive summands. Subtracting the elementary harmonic contribution makes it geometrically convergent.

Corollary 3.3 (An accelerated odd-zeta identity). *One has*

$$\boxed{\gamma_1 = 2 \log 2 - \log^2 2 - 1 + \sum_{j=2}^{\infty} \frac{\zeta'(2j-1) + H_{2j-2}(\zeta(2j-1) - 1)}{j(2j-1)}}. \quad (3.12)$$

If the sum is truncated at $j = J \geq 2$, the absolute tail is bounded by

$$\frac{16}{3} \frac{2 + \log(2J+2)}{(J+1)^2} 4^{-(J+1)}. \quad (3.13)$$

Proof. The even part of $\sum_{n \geq 0} H_n x^n = -\log(1-x)/(1-x)$ gives

$$\begin{aligned} \sum_{j=2}^{\infty} \frac{H_{2j-2}}{j(2j-1)} &= 2 \int_0^1 (1-x) \sum_{m \geq 1} H_{2m} x^{2m} dx \\ &= - \int_0^1 \log(1-x) dx - \int_0^1 \frac{1-x}{1+x} \log(1+x) dx \\ &= 2 \log 2 - \log^2 2. \end{aligned}$$

Tonelli or absolute convergence justifies the interchange. Subtract this value from (3.11).

For $p \geq 3$, integral comparison yields

$$\zeta(p) - 1 \leq 2^{1-p}, \quad |\zeta'(p)| \leq 2 \cdot 2^{-p}.$$

Together with $H_{2j-2} \leq 1 + \log(2j - 2)$, each absolute summand in the accelerated series is at most $4(2 + \log(2j))4^{-j}/j^2$. The factor $(2 + \log(2j))/j^2$ decreases, and summing the geometric tail gives (3.13). $\square$

For actual evaluation this tail bound is combined with controlled evaluation of the finitely many zeta values and derivatives. A small formal tail alone does not certify floating-point special-function evaluations. The accompanying program explicitly labels its numerical comparisons as diagnostics.

The same positivity gives useful global estimates. If $A(h) = h(\log^2 h - 2 \log h + 2 + \pi^2/3) + (2\gamma_1 - \pi^2/3)h^2$, then

$$A(h) - 2(1 + \gamma_1)h^4 \leq V(h) \leq A(h) - \frac{\zeta'(3) + \frac{3}{2}\zeta(3)}{3}h^4, \quad 0 < h < 1. \quad (3.14)$$

The left inequality is sharp at $h = 1$ by continuity, and the right has the exact first omitted coefficient at zero. For computation one may replace h by $\min(h, 1 - h)$ before using the convergent series.

4 A uniform transition at outer order one

This section proves the near-critical reversal conjecture `research:conj:crossing` in the pinned continuation `tetralogarithm-euler-polynomial-zeros` and determines its first logarithmic correction. The distinction between an expansion for fixed parameters and an expansion uniform at an integer outer order is essential: the coefficient of the algebraic density vanishes at outer order one, while an exponentially small density survives. Both contributions are of the same size at the reversal.

4.1 Definitions and the inherited sign theorem

For $a, b > 0$, write

$$F_{a,b}(z) = \sum_{n \geq 1} \frac{H_{n-1}^{(b)}}{n^a} z^n, \quad H_m^{(b)} = \sum_{j=1}^m j^{-b}, \quad g_{a,b} = \operatorname{Im} F_{a,b}(i),$$

where the boundary value is its analytic continuation. Define

$$f_n = \frac{H_{2n}^{(b)}}{(2n+1)^a}, \quad d_j = \sum_{n=0}^j (-1)^n \binom{j}{n} f_n, \quad E_N = \sum_{j=0}^{N-1} \frac{d_j}{2^{j+1}}, \quad R_N(a, b) = 2^N (E_N - g_{a,b}).$$

The axis $a = 0$ uses the same analytic convention. The comparison of interest fixes the total order:

$$\mathcal{D}_{a,b}(N) = R_N(a, b) - R_N(0, a + b).$$

Set

$$h_b(T) = \frac{1}{\Gamma(b)} \int_T^\infty \frac{v^{b-1}}{e^v - 1} dv, \quad I^a h_b(T) = \frac{1}{\Gamma(a)} \int_0^T (T - v)^{a-1} h_b(v) dv, \quad (4.1)$$

$$D_{a,b}(T) = (I^a h_b)'(T) + \frac{T^{a+b-1}}{\Gamma(a+b)(e^T - 1)}, \quad q(T) = 1 - e^{-2T}. \quad (4.2)$$

The exact compensated Euler formula yields

$$\mathcal{D}_{a,b}(s) = - \int_0^\infty D_{a,b}(T) \frac{e^{-T} q(T)^s}{1 + e^{-2T}} dT, \quad s \geq 1. \quad (4.3)$$

It agrees with the preceding definition at integers and supplies the real interpolation used below. All its derivatives in s are ordinary absolutely convergent integrals. In fact, with $\ell(T) = -\log q(T) > 0$,

$$(-1)^r \mathcal{D}_{a,b}^{(r)}(s) = - \int_0^\infty D_{a,b}(T) \frac{e^{-T} q(T)^s \ell(T)^r}{1 + e^{-2T}} dT. \quad (4.4)$$

The prior phase theorem [11] establishes the following facts. For $0 < a < 1$, $D_{a,b}(T)$ has precisely one sign change, from positive to negative. For every integer $r \geq 0$, $\mathcal{D}_{a,b}^{(r)}$ has one simple zero $\nu_r(a, b) > 1$, and

$$1 < \nu_0(a, b) < \nu_1(a, b) < \dots$$

The sign of $(-1)^r \mathcal{D}_{a,b}^{(r)}$ is negative before that zero and positive afterwards. In particular $\nu_0 = \nu$ is the reversal index, while ν_1 is the location of the unique positive maximum of $\mathcal{D}_{a,b}$. These facts can also be checked directly from (4.4): the ratio of the weights at two s values is strictly increasing in T , so subtracting its value at the density crossing proves the one-zero property. A similar comparison with the strictly decreasing function $\ell(T)$ orders the derivative zeros. We use this already proved sign theorem to identify the unique roots isolated by our uniform estimates.

Throughout the uniform statements, $B = [b_-, b_+]$ is any fixed compact interval in $(0, \infty)$ and r is a fixed nonnegative integer. Put

$$a = 1 - \varepsilon, \quad \lambda = \log(1/\varepsilon), \quad p = b + 1.$$

Choose $\varepsilon_0 < \min(1/2, b_-/2)$. In particular, the parameters used here are strictly in the finite-measure region $a + b > 1$; no unweighted hypersingular integral is introduced in this joint limit.

4.2 The reversal asymptotic and its correction

Define

$$K_{b,r} = b\Gamma(b+1)\zeta(b+1) \frac{\Gamma(r+1/2)}{\Gamma(r+1)}, \quad (4.5)$$

$$\delta_{b,r} = b - \frac{b}{2}\psi(r+1) - \frac{1}{2}\psi(r+1/2) - \frac{(b+1)\zeta(b+2)}{2\zeta(b+1)}. \quad (4.6)$$

All zeta values in these constants lie in their absolutely convergent range, and both gamma factors have positive arguments.

Theorem 4.1 (Near-critical reversal and derivative zeros). *As $\varepsilon \downarrow 0$, uniformly for $b \in B$ and each fixed r ,*

$$\begin{aligned} \frac{1}{2} \log \nu_r(1 - \varepsilon, b) &= \lambda + p \log \lambda - \log K_{b,r} \\ &\quad + \frac{p^2 \log \lambda - p \log K_{b,r} + \delta_{b,r}}{\lambda} + O_{B,r} \left(\frac{(\log \lambda)^2}{\lambda^2} \right). \end{aligned} \quad (4.7)$$

Equivalently,

$$\begin{aligned} \nu_r(1 - \varepsilon, b) &= \frac{\lambda^{2b+2}}{K_{b,r}^2 \varepsilon^2} \left[1 + \frac{2p^2 \log \lambda - 2p \log K_{b,r} + 2\delta_{b,r}}{\lambda} \right. \\ &\quad \left. + O_{B,r} \left(\frac{(\log \lambda)^2}{\lambda^2} \right) \right]. \end{aligned} \quad (4.8)$$

For $r = 0$, this proves the proposed leading law

$$\nu(1 - \varepsilon, b) \sim \frac{(\log(1/\varepsilon))^{2b+2}}{\pi b^2 \Gamma(b+1)^2 \zeta(b+1)^2 \varepsilon^2}. \quad (4.9)$$

Its first correction uses

$$\delta_{b,0} = b + \frac{b+1}{2}\gamma + \log 2 - \frac{(b+1)\zeta(b+2)}{2\zeta(b+1)}. \quad (4.10)$$

The $\log \log(1/\varepsilon)/\log(1/\varepsilon)$ correction is substantial at accessible parameter values. Keeping the implicit logarithmic equation gives a more useful approximation.

Corollary 4.2 (An approximation in terms of Lambert W). *For sufficiently small ε , put*

$$\widehat{L}_{b,r} = -p W_{-1} \left(-\frac{(K_{b,r}\varepsilon)^{1/p}}{p} \right). \quad (4.11)$$

Here W_{-1} is the real branch with values at most -1 . Then $\widehat{L}_{b,r} > p$ and, uniformly for $b \in B$,

$$\frac{1}{2} \log \nu_r(1 - \varepsilon, b) = \widehat{L}_{b,r} + \frac{\delta_{b,r}}{\widehat{L}_{b,r} - p} + O_{B,r}(\widehat{L}_{b,r}^{-2}). \quad (4.12)$$

We now prove the theorem with the estimates that control both limits.

4.3 Preserving the vanishing algebraic amplitude

The seed moments are

$$m_j(b) = \int_0^\infty v^j h_b(v) dv = \frac{(b)_{j+1}}{j+1} \zeta(b+j+1). \quad (4.13)$$

Indeed, Tonelli first integrates v^j up to the outer variable, and then expansion of $(e^v - 1)^{-1}$ gives a sum of gamma integrals. In particular,

$$m_0 = b\zeta(b+1), \quad m_1 = \frac{b(b+1)}{2} \zeta(b+2).$$

Every fixed moment is uniformly bounded as b ranges over B .

Lemma 4.3 (A uniform density estimate with an ε factor). *For some M depending only on B , uniformly for $T \geq 2$, $b \in B$, and $0 \leq \varepsilon \leq \varepsilon_0$,*

$$\begin{aligned} (I^{1-\varepsilon} h_b)'(T) &= h_b(T) - \frac{\varepsilon T^{-1-\varepsilon}}{\Gamma(1-\varepsilon)} \left[m_0 + \frac{(1+\varepsilon)m_1}{T} + O_B(T^{-2}) \right] \\ &\quad + O_B \left(\varepsilon e^{-T/2} (1+T)^M \right). \end{aligned} \quad (4.14)$$

Moreover,

$$h_b(T) = \frac{e^{-T} T^{b-1}}{\Gamma(b)} \{1 + O_B(T^{-1})\}. \quad (4.15)$$

The $O_B(T^{-2})$ term in (4.14) is inside the bracket multiplied by ε .

Proof. For $T > 0$, the regularized differentiation formula is

$$(I^{1-\varepsilon} h_b)'(T) = \frac{1}{\Gamma(1-\varepsilon)} \left[T^{-\varepsilon} h_b(T) + \varepsilon \int_0^T \frac{h_b(T) - h_b(v)}{(T-v)^{1+\varepsilon}} dv \right]. \quad (4.16)$$

The integral is absolutely convergent. At $v = 0$, $h_b(v)$ is locally integrable; at $v = T$, the numerator is $O(T - v)$ and $\varepsilon < 1$. One way to derive the identity without using an endpoint value $h_b(0)$ is to replace the fractional kernel by $(T - v + \eta)^{-\varepsilon}$, differentiate, subtract $h_b(T)$ in the differentiated integral, and let $\eta \downarrow 0$. Dominated convergence applies to the subtracted integral and to the original fractional integral on compact positive T intervals. Its boundary term tends to $T^{-\varepsilon}h_b(T)$. The same computation at $\varepsilon = 0$ gives $(I^1 h_b)' = h_b$.

Split the integral in (4.16) at $v = T/2$. On the first part retain $-h_b(v)$ and expand

$$(T - v)^{-1-\varepsilon} = T^{-1-\varepsilon} \left[1 + (1 + \varepsilon) \frac{v}{T} + O\left(\frac{v^2}{T^2}\right) \right].$$

The Taylor bound is uniform for $0 \leq v/T \leq 1/2$ and the chosen compact ε interval. Replacing the truncated moments by the full moments changes the result by an exponential error, since $h_b(v)$ decays exponentially at infinity with a uniform polynomial bound. The second moment bounds the Taylor remainder and yields the bracket in (4.14).

Here are bounds for the remaining terms. For $u \geq 1$,

$$|h_b(u)| + |h'_b(u)| \leq C_B e^{-u} (1 + u)^{M_0}.$$

For $v \geq T/2$, the mean value theorem gives

$$|h_b(T) - h_b(v)| \leq C_B e^{-T/2} (1 + T)^{M_0} \min(T - v, 1).$$

Consequently its integral against $(T - v)^{-1-\varepsilon}$ is at most $C_B e^{-T/2} (1 + T)^{M_0+1}$, uniformly in ε : the integral on $0 < T - v < 1$ is bounded by $(1 - \varepsilon_0)^{-1}$, and the rest by $1 + \log T$. The $h_b(T)$ contribution on $v < T/2$ satisfies the same type of bound. Finally

$$\left| \frac{T^{-\varepsilon}}{\Gamma(1 - \varepsilon)} - 1 \right| \leq C\varepsilon(1 + \log T),$$

by the mean value theorem in ε . Multiplication by $h_b(T)$ absorbs this discrepancy in the exponential error. Every discarded term in this paragraph carries the prefactor ε from (4.16), or the same factor from the last inequality. Increasing M proves (4.14).

For (4.15), expand the Bose denominator as $e^{-v}\{1 + O(e^{-v})\}$. In the leading integral set $v = T + u$, factor $e^{-T}T^{b-1}$, and use $(1 + u/T)^{b-1} = 1 + O_B(u(1 + u)^{M_1}/T)$ inside its exponential integral. This proves the estimate uniformly on B . $\square$

4.4 A joint expansion under the Euler kernel

Put

$$A_{b,r} = \frac{m_0(b)\Gamma(r + 1/2)}{2}, \quad B_{b,r} = \frac{\Gamma(r + 1)}{2\Gamma(b + 1)}, \quad (4.17)$$

$$\alpha_{b,r} = \frac{1}{2}\psi(r + 1/2) + \frac{m_1(b)}{m_0(b)}, \quad \beta_{b,r} = b - \frac{b}{2}\psi(r + 1). \quad (4.18)$$

Thus $K_{b,r} = A_{b,r}/B_{b,r}$ and $\delta_{b,r} = \beta_{b,r} - \alpha_{b,r}$.

Lemma 4.4 (Uniform competition of two scales). *Let $L = \frac{1}{2} \log s$ and $\lambda = \log(1/\varepsilon)$. Uniformly for $b \in B$ and $\lambda/2 \leq L \leq 2\lambda$, as $\varepsilon \downarrow 0$,*

$$\begin{aligned} (-1)^r \mathcal{D}_{1-\varepsilon,b}^{(r)}(s) &= \varepsilon A_{b,r} s^{-r-1/2} L^{-1} \{1 + \alpha_{b,r}/L + O_{B,r}(L^{-2})\} \\ &\quad - B_{b,r} s^{-r-1} L^b \{1 + \beta_{b,r}/L + O_{B,r}(L^{-2})\}. \end{aligned} \quad (4.19)$$

The two remainder terms are bounded separately by their respective displayed scales. In particular the estimate remains useful where the leading terms cancel.

Proof. Insert Lemma 4.3 into (4.2) and (4.4). The change of variable $t = \sqrt{s}e^{-T}$ gives $T = L - \log t$ and $e^{-T} dT = -s^{-1/2} dt$. On the central interval $s^{-1/4} \leq t \leq s^{1/4}$ one has $T \in [L/2, 3L/2]$ and

$$\frac{(1 - t^2/s)^s}{1 + t^2/s} \longrightarrow e^{-t^2},$$

$$\{s[-\log(1 - t^2/s)]\}^r \longrightarrow t^{2r}.$$

Their differences from the limits have integrals bounded by $O_r(s^{-1})$ times fixed Gaussian moments. For example, when $t^2/s \leq 1/2$,

$$\left| \frac{(1 - t^2/s)^s}{1 + t^2/s} - e^{-t^2} \right| \leq C s^{-1} e^{-t^2} (t^2 + t^4),$$

and the second replacement has an analogous bound with a further fixed polynomial in t . The powers of T remain comparable to their powers of L on this central interval.

The tails require no uniformity assumption beyond the stated window. For $T < L/2$, use $q(T)^s \leq \exp(-se^{-2T}) \leq \exp(-e^L)$. The local bounds can be made explicit: for $0 < T \leq 1$,

$$|D_{1-\varepsilon, b}(T)| \leq C_B(T^{-\varepsilon_0} + T^{b-1-\varepsilon_0})(1 + |\log T|).$$

Indeed $h_b(T) \leq C_B(1 + T^{b-1})(1 + |\log T|)$ follows by splitting its integral at one; insertion in the compensated derivative formula, or its beta-integral form (4.25), gives the displayed estimate. Since $\varepsilon_0 < b_-/2$ and $\varepsilon_0 < 1/2$, it is integrable even after multiplication by the additional fixed powers of $\log T$ coming from $\ell(T)^r$. For $T \geq 1$ the density lemma gives a common polynomial bound. This entire contribution is smaller than either displayed scale times every power of L^{-1} ; the window implies $e^{-2L} \leq \varepsilon \leq e^{-L/2}$. For $T > 3L/2$, equivalently $0 < t < s^{-1/4}$, each algebraic contribution retains its factor ε and is bounded by the elementary moments of t^{2r} over this shrinking interval. Exponential contributions have the extra factor t and are bounded still more strongly. Logarithmic factors are harmless, since all fixed moments of $t^c |\log t|^j$ at zero are finite for $c > -1$.

For clarity, the exponential error in (4.14) contributes at most

$$O_{B,r}(\varepsilon s^{-r-3/4}(1+L)^M).$$

Its ratio to the algebraic scale is $O_{B,r}(s^{-1/4}(1+L)^{M+1})$, which is smaller than L^{-2} uniformly in the window. The $O(e^{-2T})$ terms produced by the Bose denominator are likewise negligible relative to the appropriate exponential scale. These estimates explain why simply deleting an exponential density in the fixed-parameter expansion would be invalid.

It remains to compute the first corrections. Uniformly in the window, $\varepsilon \log L = o(L^{-k})$ for every fixed k . Thus $L^{-\varepsilon}$, $\Gamma(1-\varepsilon)^{-1}$, and $\Gamma(b+1-\varepsilon)^{-1}$ may be replaced by their values at $\varepsilon = 0$ at the required relative accuracy. For any exponent c in a fixed compact real set, Taylor's theorem on $|\log t| \leq L/2$ gives

$$(L - \log t)^c = L^c \left[1 - \frac{c \log t}{L} + O\left(\frac{(\log t)^2}{L^2}\right) \right].$$

The full moments needed after extending the central interval are

$$\int_0^\infty t^{2r} e^{-t^2} dt = \frac{1}{2} \Gamma(r + 1/2), \quad \frac{\int_0^\infty t^{2r} e^{-t^2} \log t dt}{\int_0^\infty t^{2r} e^{-t^2} dt} = \frac{1}{2} \psi(r + 1/2),$$

$$\int_0^\infty t^{2r+1} e^{-t^2} dt = \frac{1}{2} \Gamma(r + 1), \quad \frac{\int_0^\infty t^{2r+1} e^{-t^2} \log t dt}{\int_0^\infty t^{2r+1} e^{-t^2} dt} = \frac{1}{2} \psi(r + 1).$$

The algebraic density consequently gives $A_{b,r}$ and $\alpha_{b,r}$. The dominant positive density $e^{-T} T^b / \Gamma(b+1)$ gives the correction $-b\psi(r+1)/2$; the retained seed $h_b(T)$ gives the additional b/L , because $\Gamma(b+1)/\Gamma(b) = b$. Hence its coefficient is $\beta_{b,r}$. The separately bounded second logarithmic moments prove both remainders in (4.19). $\square$

4.5 Locating and expanding the unique zero

Proof of Theorem 4.1. Define

$$S_{\varepsilon,b,r} = \frac{\lambda^{2p}}{K_{b,r}^2 \varepsilon^2}.$$

For $s = xS_{\varepsilon,b,r}$ and x in any fixed compact interval of $(0, \infty)$, $L = \lambda + p \log \lambda - \log K_{b,r} + \frac{1}{2} \log x$, so $L/\lambda \rightarrow 1$ uniformly for $b \in B$. In (4.19), the ratio of the first positive scale to the second positive scale is

$$\varepsilon K_{b,r} \sqrt{s} L^{-p} = \sqrt{x} (\lambda/L)^p \rightarrow \sqrt{x}.$$

Thus the comparison has opposite signs at any fixed $x < 1$ and fixed $x > 1$, uniformly for $b \in B$. The inherited uniqueness theorem identifies its zero and proves

$$\nu_r(1 - \varepsilon, b)/S_{\varepsilon,b,r} \rightarrow 1.$$

This first localization justifies using the same window at the zero; the window has not been assumed as an unproved property of ν_r .

At the zero set $L_r = \frac{1}{2} \log \nu_r$. Equating the two terms of (4.19) and dividing their positive factors gives

$$\varepsilon K_{b,r} e^{L_r} L_r^{-p} = 1 + \frac{\delta_{b,r}}{L_r} + O_{B,r}(L_r^{-2}).$$

Taking logarithms yields the uniform equation

$$L_r = \lambda + p \log L_r - \log K_{b,r} + \frac{\delta_{b,r}}{L_r} + O_{B,r}(L_r^{-2}). \quad (4.20)$$

First $L_r = \lambda + O_B(\log \lambda)$, and substitution once more gives $L_r = \lambda + p \log \lambda - \log K_{b,r} + O_{B,r}(\log \lambda/\lambda)$. Expand $\log L_r$ about λ in (4.20). The result is (4.7); Taylor's remainder is $O_{B,r}((\log \lambda)^2/\lambda^2)$. Exponentiating gives (4.8). The cases $r = 0$ follow from $\Gamma(1/2) = \sqrt{\pi}$, $\psi(1) = -\gamma$, and $\psi(1/2) = -\gamma - 2 \log 2$. $\square$

Proof of Corollary 4.2. The number $\hat{L} = \hat{L}_{b,r}$ in (4.11) is the large solution of

$$\hat{L} - p \log \hat{L} = \lambda - \log K_{b,r}.$$

It satisfies $\hat{L} \sim \lambda$, uniformly on B . On that range the derivative of $L - p \log L$ is $1 - p/L$, bounded away from zero. Comparing with (4.20) first gives $L_r - \hat{L} = O_{B,r}(\hat{L}^{-1})$. One Taylor expansion then gives

$$(1 - p/\hat{L})(L_r - \hat{L}) = \delta_{b,r}/\hat{L} + O_{B,r}(\hat{L}^{-2}),$$

which proves (4.12). $\square$

4.6 A universal profile and the positive maximum

For $r = 0$, abbreviate $K_b = K_{b,0}$ and set

$$S_{\varepsilon,b} = \frac{\lambda^{2b+2}}{K_b^2 \varepsilon^2}, \quad M_{\varepsilon,b} = \frac{\pi b^2 \Gamma(b+1) \zeta(b+1)^2}{2} \frac{\varepsilon^2}{\lambda^{b+2}}. \quad (4.21)$$

Theorem 4.5 (Universal profile and derivative-root ratios). *For every compact $X \subset (0, \infty)$ and every fixed nonnegative integer m ,*

$$\frac{\mathcal{D}_{1-\varepsilon,b}(S_{\varepsilon,b}x)}{M_{\varepsilon,b}} \rightarrow x^{-1/2} - x^{-1} \quad \text{in } C^m(X), \quad (4.22)$$

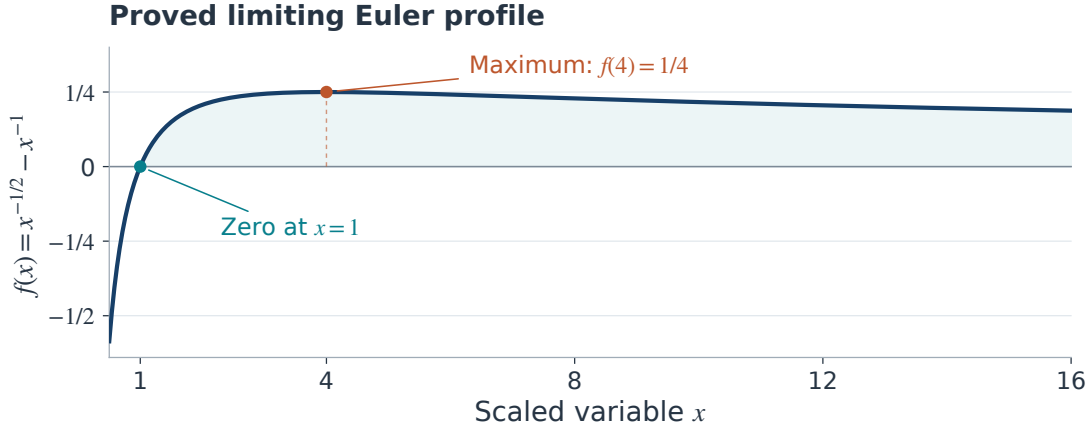


Figure 2: The proved limiting profile $x^{-1/2} - x^{-1}$ from Theorem 4.5, shown for $1/2 \leq x \leq 16$. Its zero at one locates the reversal scale and its maximum at four locates the later positive maximum. This is the limiting function, not a finite- ε numerical comparison.

uniformly for $b \in B$. Consequently, for every fixed r ,

$$\frac{\nu_r(1 - \varepsilon, b)}{\nu_0(1 - \varepsilon, b)} \longrightarrow \left\{ \frac{\sqrt{\pi}\Gamma(r+1)}{\Gamma(r+1/2)} \right\}^2 = \left\{ \frac{4^r}{\binom{2r}{r}} \right\}^2. \quad (4.23)$$

In particular the maximum is attained at an index asymptotic to $4\nu_0$, and its height satisfies

$$\max_{s \geq 1} \mathcal{D}_{1-\varepsilon, b}(s) \sim \frac{\pi b^2 \Gamma(b+1) \zeta(b+1)^2}{8} \frac{\varepsilon^2}{\lambda^{b+2}}. \quad (4.24)$$

The same height asymptotic holds when the maximum is restricted to integer truncation orders. Any maximizing integer is one of the two integers adjacent to ν_1 . The first integer with a strictly positive comparison is $\lfloor \nu_0 \rfloor + 1$ and therefore has the same asymptotic as (4.9).

Proof. The two scales in (4.19), evaluated at $s = S_{\varepsilon, b}x$, both equal $M_{\varepsilon, b}$ at $x = 1$ to leading order. Their x powers are $x^{-1/2}$ and x^{-1} . For every $r \leq m$, the already justified differentiated integral in Lemma 4.4 gives

$$\frac{S_{\varepsilon, b}^r \mathcal{D}_{1-\varepsilon, b}^{(r)}(S_{\varepsilon, b}x)}{M_{\varepsilon, b}} \longrightarrow (-1)^r \left[(1/2)_r x^{-r-1/2} - r! x^{-r-1} \right]$$

uniformly on X . This proves C^m convergence directly, rather than by differentiating an undifferentiated asymptotic. The root ratios follow either from these limiting derivatives and uniqueness or from (4.8). The gamma duplication formula gives the central-binomial expression.

The limiting profile has its unique positive maximum at $x = 4$, with value $1/4$. The inherited global one-maximum theorem places the actual maximum at ν_1 , and (4.23) puts $\nu_1/S_{\varepsilon, b} \rightarrow 4$. Evaluation of the uniform profile there gives (4.24). No global maximum is inferred solely from convergence on compact x intervals. Strict increase before ν_1 and strict decrease afterwards reduce the integer maximum to the adjacent integers; their rescaled locations tend to four as well. The asserted first positive integer follows from the strict sign classification at ν_0 . $\square$

b	ε	$\frac{1}{2} \log \nu_0$ (quadrature)	Error of (4.12)
1	10^{-6}	18.6740498534	$-9.5251 \cdot 10^{-3}$
1	10^{-12}	33.6353818511	$-2.7315 \cdot 10^{-3}$
1	10^{-24}	62.4857641121	$-7.6396 \cdot 10^{-4}$
2	10^{-6}	20.8878708711	$-1.0288 \cdot 10^{-2}$
2	10^{-12}	36.3240064653	$-2.9060 \cdot 10^{-3}$
2	10^{-24}	65.7078545360	$-8.1393 \cdot 10^{-4}$

Table 1: Direct exact-kernel quadrature and the Lambert approximation for $r = 0$. The last column is the numerical value minus the explicit approximation with its $O(\widehat{L}^{-2})$ term omitted. These are numerical diagnostics. The asymptotic theorem does not attach an effective finite- ε bound to this column.

4.7 Verification, integration status, and further questions

The numerical verifier evaluates the exact density through the cancellation-free identity

$$D_{1-\varepsilon,b}(T) = \frac{T^{-\varepsilon}}{\Gamma(1-\varepsilon)} \left[h_b(T) - \frac{1}{\Gamma(b)} \int_0^T \frac{v^{b-1} \{(1-v/T)^{-\varepsilon} - 1\}}{e^v - 1} dv \right] + \frac{T^{b-\varepsilon}}{\Gamma(b+1-\varepsilon)(e^T - 1)}. \quad (4.25)$$

It uses `expm1` and `log1p` for the small numerator. This expression follows by interchanging integrals in $I^a h_b$ before differentiation and is independent of the truncated large- T expansion used in the proof. At $b = 1$ it uses $h_1(T) = -\log(1 - e^{-T})$; at $b = 2$ it uses $h_2(T) = \text{Li}_2(e^{-T}) - T \log(1 - e^{-T})$. The outer integral is evaluated after the exact Gaussian change of variables. The output checks $r = 0, 1, 2$, several small values of ε , and a tighter-tolerance repeat. These values are floating-point diagnostics, not certified enclosures or proof premises.

The named conjecture `research:conj:crossing` can now be promoted to a theorem, with (4.8) as a stronger replacement. The source’s fixed-parameter fractional theorem and one-crossing theorem remain valid; the present result supplies the missing uniform estimate. Its role is specifically the joint regime $a \uparrow 1$, $s \rightarrow \infty$. The compactness condition on b is part of the theorem. It does not cover $b \downarrow 0$, $b \rightarrow \infty$, or derivative orders r growing with λ .

Several next questions are now concrete. First, expand the bracket in (4.14) to arbitrary order and retain enough terms of h_b to obtain a full uniform logarithmic expansion of ν_r . Its coefficients should be expressible through seed zeta moments and polygamma values at $r + 1/2$ and $r + 1$; the displayed proof identifies the required coefficient-generating operations. Second, derive explicit finite constants in the two separate remainders of (4.19), so that the Lambert approximation becomes a practical certified locator. Third, study the simultaneous limit $r \rightarrow \infty$, $\varepsilon \downarrow 0$. The fixed- r ratio in (4.23) grows like πr , but exchanging these limits is not justified here. Fourth, the limits $b \downarrow 0$ and $b \rightarrow \infty$ alter the seed moments and may require different transition variables. Finally, extend the two-scale mechanism to Euler kernels at other roots of unity, where a scalar sign-change theorem may need a sector-valued substitute.

5 Sharp sign blocks for diagonal Lerch velocities

We use the notation and theorems of the latest Euler–Lambert continuation [12]. To make the connection with the original special functions explicit, define for $h > 0$, $0 \leq \rho \leq 1$,

$$f_{n,k}(x) = (-1)^{n+k} \frac{d^k}{dx^k} \frac{\log^n x}{x}, \quad \mathcal{F}_{n,k}^{\rho,h}(x) = \sum_{m \geq 0} \rho^m f_{n,k}(x + mh).$$

For $k \geq 1$ this series converges locally uniformly for $x > 0$ even at $\rho = 1$. At $h = 1$, it equals $(-1)^{n+k}$ times the k th parameter derivative of the logarithmic Lerch coefficient $(-1)^n \partial_s^n \Phi(\rho, s, x)|_{s=1}$ when $\rho < 1$, and $(-1)^{n+k} \gamma_n^{(k)}(x)$ when $\rho = 1$. At $n \geq 2$, $k = n - 1$, the local expansion $\log^n x/x = (x-1)^n + O((x-1)^{n+1})$ gives $f_{n,n-1}(1) = 0$ and $f'_{n,n-1}(1) = -n!$. The implicit function theorem therefore gives a real analytic zero branch $a_n^{(h)}(\rho)$ with $a_n^{(h)}(0) = 1$. Differentiating its equation at zero proves

$$(a_n^{(h)})'(0) = \frac{f_{n,n-1}(1+h)}{n!}.$$

The real-rooted elementary polynomial argument in the cited baseline also shows that this is the smallest positive zero branch for sufficiently small $\rho \geq 0$.

For $A = 1 + h > 1$, put

$$v_n(A) = -\frac{1}{n!} \frac{d^{n-1}}{dx^{n-1}} \frac{\log^n x}{x} \Big|_{x=A}, \quad \sum_{n \geq 1} v_n(A) t^n = \log A - U_A(t), \quad e^{U_A(t)} = A + tU_A(t).$$

Thus $v_n(A)$ is exactly that initial velocity. The auxiliary coefficient at $n = 1$ completes the generating function. For completeness, apply Lagrange inversion [7] to $X = A + t \log X$ with outer function $\log X$. It gives

$$[t^n] \log X(t) = \frac{1}{n!} \frac{d^{n-1}}{dx^{n-1}} \frac{\log^n x}{x} \Big|_{x=A} = -v_n(A).$$

Setting $U_A = \log X$ proves the displayed inverse identity as an identity of analytic germs. Equivalently, with unsigned Stirling numbers of the first kind,

$$v_n(A) = A^{-n} \sum_{j=1}^n \left[\begin{matrix} n \\ n-j+1 \end{matrix} \right] \frac{(-\log A)^j}{j!}.$$

These generating and coefficient identities are inherited results, not new claims of this section. Define $\theta \in (0, \pi)$, a , u_* , τ , R , and χ by

$$\begin{aligned} \log A &= 1 - \theta \cot \theta + \log \frac{\theta}{\sin \theta}, \\ a &= 1 - \theta \cot \theta, \quad u_* = a + i\theta, \quad \tau = e^{u_*}, \quad R = e^a, \quad \chi = \sqrt{-2u_*}, \end{aligned}$$

where $\Re \chi > 0$ and $\Im \chi < 0$. Write $\phi = -\arg \chi \in (\pi/4, \pi/2)$. The selected-sheet theorem in the source report proves the additive estimate

$$\frac{\sqrt{\pi} R^n n^{3/2}}{|\chi|} v_n(A) = \cos(n\theta + \phi) + O_A(n^{-1}). \quad (5.1)$$

No new inverse-sheet claim is needed here.

Theorem 5.1 (The least eventual block length). *For each fixed $A > 1$, the least positive integer K such that every sufficiently late block of K consecutive velocities contains both a strictly positive and a strictly negative velocity is*

$$K(A) = \left\lceil \frac{\pi}{\theta(A)} \right\rceil + 1. \quad (5.2)$$

In particular, $K(2) = 4$, and infinitely many blocks of three consecutive velocities $v_n(2)$ have one strict sign.

Proof. First suppose π/θ is not an integer, and put $m = \lfloor \pi/\theta \rfloor$. The $m+2$ points $0, \theta, \dots, (m+1)\theta$ have largest circular gap

$$g = \max\{\theta, 2\pi - (m+1)\theta\} < \pi.$$

There is consequently a fixed $\eta > 0$ such that every rotation of these points contains one point with cosine at least η and one with cosine at most $-\eta$. Indeed choose an arc length strictly between g and π , and take the two opposite closed arcs of that length inside the positive and negative cosine semicircles. A gap bound forces each arc to meet the rotated point set. The error in (5.1) is eventually smaller than $\eta/2$. Thus $m + 2 = K(A)$ always suffices.

To prove sharpness, $m + 1$ consecutive phase points all lie in an open cosine semicircle precisely for starting phases in either of two antipodal open intervals, each of length $\delta = \pi - m\theta > 0$. If $\theta/(2\pi)$ is irrational, density of the rotation orbit supplies infinitely many starting phases in a closed subinterval of one of these intervals. They have a uniform nonzero cosine margin, so (5.1) gives infinitely many strict same-sign velocity blocks of length $m + 1$.

Now suppose $\theta/(2\pi) = p/q$ in lowest terms. The finite phase orbit modulo π has spacing π/q when q is odd, and spacing $2\pi/q$ when q is even. The two antipodal admissible intervals project to one open interval modulo π of length

$$\delta = \frac{\pi(q - 2mp)}{q}.$$

The positive integer $q - 2mp$ is odd when q is odd, and even when q is even. Thus δ is at least one spacing in either case. No point of the phase orbit can be an endpoint of this interval: an endpoint would make one of the finitely many cosines zero. We recall the elementary exclusion. If such a cosine vanished, then ϕ/π would be rational, so $b = \theta/a = \tan(\pi - 2\phi) > 0$ would be algebraic. Since $\cot \theta$ is also algebraic, the identity

$$\theta(1 + b \cot \theta) = b$$

would make θ algebraic; its coefficient on the left cannot vanish because $b > 0$. This contradicts the transcendence of π . The interval therefore contains a phase point strictly inside. Its residue class gives infinitely many same-sign blocks, and its positive cosine margin again absorbs the error. This proves sharpness in the rational case as well.

Finally suppose $\theta = \pi/m$ for an integer $m \geq 2$. The phase orbit is rational and, by the preceding exclusion, has no zero cosine. In every block of $m + 1$ phases, the first and last are antipodal, so both strict signs occur, with a uniform margin. Blocks of m phases of one sign also occur: their admissible interval modulo π has length θ , exactly the orbit spacing, and its endpoints are excluded. Thus $K = m + 1$, as stated. For $A = 2$, the already proved bounds $\pi/3 < \theta(2) < \pi/2$ give $K(2) = 4$. $\square$

This answers the least-block-length question in the newest report. It does not decide whether $\theta(2)/(2\pi)$ is irrational. In the exceptional reciprocal-angle case, the formula also improves the earlier sufficient bound by one.

6 Higher critically weighted sums and their finite constants

The selected branch has a convergent Puiseux expansion at τ ,

$$U_A(t) = u_* + \sum_{m \geq 1} c_m (1 - t/\tau)^{m/2}, \quad c_1 = \chi. \quad (6.1)$$

The first coefficients, with $U = u_*$, are

$$\begin{aligned} c_2 &= \frac{U}{3} - 1, \\ c_3 &= \chi \left(\frac{1}{2} - \frac{U}{18} - \frac{1}{4U} \right), \\ c_4 &= -\frac{2U^2}{135} + \frac{U}{3} - \frac{2}{3}. \end{aligned} \quad (6.2)$$

The first three are in the source; the fourth follows from the coefficient of $z^{5/2}$ in $e^w - 1 - w + z(U + w) = 0$.

For integers $p \geq 1$ consider the sharp partial sums

$$S_p(N) = \sum_{n=1}^N n^p v_n(A) \tau^n.$$

For $p \geq 2$ these sums contain divergent oscillatory terms arising from the conjugate singularity. Subtracting only nonoscillatory fractional powers does not give a finite limit. The following explicit formula includes both contributions.

Put $q = \bar{\tau}/\tau = e^{-2i\theta} \neq 1$ and

$$B_N(\beta) = (-1)^N \binom{\beta}{N} = \frac{\Gamma(N - \beta)}{\Gamma(-\beta)\Gamma(N + 1)} \quad (\beta \notin \{0, 1, 2, \dots\}).$$

Let $\{l^p\}$ be a Stirling number of the second kind and $(\alpha)_l$ a falling factorial. Define the following finite explicit sums, with $\alpha_j = j + 1/2$:

$$T_p(N) = - \sum_{j=0}^{p-1} c_{2j+1} \sum_{l=j+1}^p \left\{ \begin{matrix} p \\ l \end{matrix} \right\} (-1)^l (\alpha_j)_l \sum_{i=0}^{l-j-1} (-1)^i \binom{l}{i} B_N(\alpha_j - l + i - 1), \quad (6.3)$$

$$\begin{aligned} E_p(N) = & -q^{-N} \sum_{j=0}^{p-2} \overline{c_{2j+1}} \sum_{l=j+2}^p \left\{ \begin{matrix} p \\ l \end{matrix} \right\} (-1)^l (\alpha_j)_l \\ & \cdot \sum_{\substack{i, h \geq 0 \\ i+h \leq l-j-2}} (-1)^{i+h} \binom{l}{i} \frac{q^h}{(1-q)^{h+1}} B_N(\alpha_j - l + i + h). \end{aligned} \quad (6.4)$$

An empty sum is zero, so $E_1 = 0$.

Theorem 6.1 (All-order sharp-cutoff finite parts). *For each fixed $A > 1$ and integer $p \geq 1$,*

$$S_p(N) = T_p(N) + E_p(N) + C_p(u_*) + O_{A,p}(N^{-1/2}), \quad (6.5)$$

where the finite constant is the universal rational polynomial

$$C_p(U) = \sum_{j=1}^p (-1)^{j+1} j! \left\{ \begin{matrix} p \\ j \end{matrix} \right\} c_{2j}(U). \quad (6.6)$$

The even Puiseux coefficients, and hence the constant, have the explicit coefficient-extraction formula

$$c_{2j}(U) = \frac{(-1)^j}{2j} [w^{2j-1}] \left(\frac{(U+w)w^2}{e^w - 1 - w} \right)^j. \quad (6.7)$$

In particular, c_{2j} and C_p have rational coefficients and degrees at most j and p , respectively. The gamma ratios in T_p, E_p can equivalently be expanded and truncated through every nonnegative power of N ; only strictly positive half-integer powers occur, and the limit in (6.5) is unchanged.

Proof. Let $F(r) = \log A - U_A(\tau r)$ and $\Theta = r d/dr$. Exact coefficient extraction gives

$$S_p(N) = [r^N] \frac{\Theta^p F(r)}{1 - r}. \quad (6.8)$$

The known slit continuation places its only dominant singularities at $r = 1$ and $r = q$. Use

$$\Theta^p = \sum_{l=1}^p \left\{ \begin{matrix} p \\ l \end{matrix} \right\} r^l \frac{d^l}{dr^l}.$$

Near $r = 1$, write $z = 1 - r$. Applying this differential operator to $-c_{2j+1}z^{\alpha_j}$ and dividing by z produces the powers

$$-c_{2j+1} \left\{ \begin{matrix} p \\ l \end{matrix} \right\} (-1)^{l+i} (\alpha_j)_l \binom{l}{i} z^{\alpha_j - l + i - 1}.$$

Precisely the exponents at most $-3/2$ produce divergent coefficient terms. Their ranges are $0 \leq j \leq p-1$, $j+1 \leq l \leq p$, and $0 \leq i \leq l-j-1$; their coefficients are exactly (6.3). The analytic part of F is $\log A - U - \sum_{j \geq 1} c_{2j} z^j$. Its contribution to the simple pole after division by z is

$$(\Theta^p F_{\text{analytic}})(1) = \sum_{j=1}^p (-1)^{j+1} j! \left\{ \begin{matrix} p \\ j \end{matrix} \right\} c_{2j} = C_p(U).$$

The remaining nonintegral exponents are at least $-1/2$.

At $r = q$, write $w = 1 - r/q$. The same differential operator has the same expression in this coordinate, while

$$\frac{1}{1-r} = \frac{1}{1-q+qw} = \sum_{h \geq 0} \frac{(-1)^h q^h}{(1-q)^{h+1}} w^h.$$

Combining this with the conjugate Puiseux series gives exponents $\alpha_j - l + i + h$. Those at most $-3/2$ are precisely the terms of (6.4). At the conjugate point there is no additional simple pole: the denominator is nonzero and the analytic part stays analytic. The remaining nonintegral exponents are again at least $-1/2$.

After subtracting the finitely many displayed singular terms and $C_p(U)/(1-r)$, singularity transfer [6] gives coefficients $O_{A,p}(N^{-1/2})$. The hypotheses are exactly the proved radial-slit continuation and convergent local Puiseux expansions; the usual finite-singularity form of transfer applies. The exact binomial coefficient identity gives (6.5). Its gamma-ratio expansion proves the equivalent fractional-power subtraction statement.

Finally the local implicit equation is $z = -(e^w - 1 - w)/(U + w)$. In the source's Lagrange formula

$$c_m = \frac{\chi^m}{m} [w^{m-1}] \left(\frac{2U(e^w - 1 - w)}{w^2(U + w)} \right)^{-m/2},$$

substitute $m = 2j$ and $\chi^{2j} = (-2U)^j$. This gives (6.7); its polynomial assertion follows because $w^2/(e^w - 1 - w)$ is analytic with rational Taylor coefficients. $\square$

Corollary 6.2 (An explicit second-derivative identity). *For the same parameters,*

$$\begin{aligned} \sum_{n=1}^N n^2 v_n(A) \tau^n &= \frac{\chi}{3\sqrt{\pi}} N^{3/2} \\ &\quad + \frac{\sqrt{N}}{\sqrt{\pi}} \left(\frac{5\chi}{8} - \frac{3c_3}{2} + \frac{\bar{\chi} e^{2iN\theta}}{2(1 - e^{-2i\theta})} \right) \\ &\quad + \frac{4u_*^2}{135} - \frac{u_*}{3} + \frac{1}{3} + O_A(N^{-1/2}). \end{aligned} \tag{6.9}$$

Proof. The finite sums simplify to

$$T_2(N) = \frac{\chi}{4} B_N(-5/2) - \frac{3c_3}{4} B_N(-3/2), \quad E_2(N) = \frac{\bar{\chi} q^{-N}}{4(1-q)} B_N(-3/2).$$

Use $B_N(-5/2) = \frac{4}{3\sqrt{\pi}} N^{3/2} (1 + 15/(8N) + O(N^{-2}))$ and $B_N(-3/2) = \frac{2}{\sqrt{\pi}} N^{1/2} (1 + O(N^{-1}))$. The constant is $c_2 - 2c_4$, giving the stated polynomial. $\square$

For $p = 1$, the theorem recovers the established formula $\sum_{n \leq N} nv_n(A)\tau^n = \chi\sqrt{N/\pi} + u_*/3 - 1 + O_A(N^{-1/2})$. The new second-order term displays the obstruction at higher orders: the conjugate critical point contributes a rotating term of size $\sqrt{N}$. Higher p produce the analogous finite collection of rotating half-integer powers, with an explicitly defined finite constant after subtraction.

Provenance and scope. These two results answer the least-block-length and higher-critical-sum questions in `sections/10_research_agenda.tex` of the Euler–Lambert continuation [12] at ProveIt commit 570b0567f311cf1890865065896be2665f469e4f. They use, and do not re-label as new, the selected-sheet theorem, all-order Puiseux expansion and additive coefficient asymptotics in that continuation’s `sections/05_harmonic.tex`. The word “new” refers to the checked repository baseline; no worldwide priority assertion follows from that comparison.

7 A Cayley normal form that preserves series depth

The inspected continuation already supplies a constructive projector for the full shuffle ideal generated by convergent Cayley identities [17]. Its image is fixed by the Cayley involution, but its expansion can increase the number of nonzero letters. We construct another projector for the same ideal whose output never has greater series depth than its input. The choice of an oriented representative, and a different equivariant endpoint projection, make this possible. We then compute the entire depth filtration of that formal quotient. These are statements about a specified algebra of words. The period map may satisfy further identities, so no numerical independence assertion is involved.

The underlying shuffle-polynomial theorem, Eulerian logarithm, and Witt character formula are classical [14, 15, 16]. The contribution here is their simultaneous compatibility with the two letter counts exchanged by the Cayley transformation, the resulting non-increasing-depth retraction, and the explicit filtered dimensions.

7.1 Alphabet, admissibility, and the two counts

Use outer-letter-first iterated integrals on $[0, 1]$ and the forms

$$z = \frac{dt}{t}, \quad c = \frac{dt}{1-t}, \quad a = \frac{idt}{1-it}, \quad b = -\frac{dt}{1+t}, \quad \bar{a} = -\frac{idt}{1+it}.$$

An ordinary convergent word has first letter different from c and last letter different from z . Introduce the adapted letters

$$x = c - b, \quad y = a - \bar{a}, \quad h = a + \bar{a} - b, \quad \mathcal{E} = \{z, y, h, b, x\}. \quad (7.1)$$

They form a rational basis of the same five-dimensional space of forms. In particular,

$$c = x + b, \quad a = \frac{y + h + b}{2}, \quad \bar{a} = \frac{-y + h + b}{2}.$$

Every adapted letter other than z is a linear combination of nonzero original letters. Hence the original series-depth filtration is exactly

$$\text{dep}(w) = |w| - n_z(w), \quad F_D\mathcal{A} = \text{span}_{\mathbb{Q}}\{w : \text{dep}(w) \leq D\}, \quad (7.2)$$

where $n_z(w)$ counts occurrences of z . Let $\tilde{\mathcal{A}}$ be the full shuffle algebra on $\mathcal{E}$, with empty word 1 and product Π . Let $\mathcal{A}$ be its subalgebra spanned by 1 and the adapted words satisfying

$$w_1 \neq x, \quad w_{|w|} \neq z. \quad (7.3)$$

This is precisely the original convergent word space: x is the only adapted letter with a c component, and the other four adapted letters span the original first-letter subspace that excludes c .

The Cayley operator and complex conjugation are

$$C(v_1 \cdots v_n) = \tau(v_n) \cdots \tau(v_1), \quad \tau z = x, \quad \tau x = z, \quad \tau y = y, \quad \tau h = -h, \quad \tau b = b, \quad (7.4)$$

$$K(w) = (-1)^{n_y(w)} w. \quad (7.5)$$

They are commuting involutive shuffle-algebra automorphisms preserving $\mathcal{A}$. The substitution $t \mapsto (1-t)/(1+t)$ has $\tau = -\phi^*$; its orientation reversal is already included in (7.4). Thus, for the convergent period map H ,

$$H(Cf) = H(f), \quad H(f \amalg g) = H(f)H(g), \quad f, g \in \mathcal{A}. \quad (7.6)$$

For example, $H(y) = i\pi/2$, $H(b) = -\log 2$, and $H(h) = 0$. The first and last exclusions in (7.3) are exchanged by reversal and $z \leftrightarrow x$, so every transformed word remains convergent.

We impose exactly the full Cayley shuffle ideal

$$\mathcal{J} = \langle f - Cf : f \in \mathcal{A} \rangle_{\amalg}, \quad \mathcal{B} = \mathcal{A}/\mathcal{J}. \quad (7.7)$$

No stuffle, distribution, or independent period relation is included in this definition. Equation (7.6) proves that the period map annihilates $\mathcal{J}$.

For an adapted word use the bidegree

$$\deg_{z,x}(w) = (n_z(w), n_x(w)).$$

Shuffle multiplication adds this bidegree. The operator C exchanges its two components, and K preserves both.

7.2 A count-preserving endpoint retraction

Lemma 7.1 (Homogeneous endpoint retraction). *There is a unique shuffle-algebra retraction $R_0 : \tilde{\mathcal{A}} \rightarrow \mathcal{A}$ fixing $\mathcal{A}$ and sending the shuffle-polynomial variables z, x to zero. It commutes with C, K and preserves the bidegree (n_z, n_x) in every nonzero term. If a word w has l initial x 's and r terminal z 's, then*

$$R_0(w) = \sum_{u=0}^l \sum_{v=0}^r (-1)^{u+v} x^u \amalg w[u : |w| - v] \amalg z^v. \quad (7.8)$$

The powers and the middle subword in this formula use concatenation.

Proof. Order the adapted alphabet as $z < y < h < b < x$. Every Lyndon word of length at least two is admissible. Indeed it cannot start with the largest letter x , and cannot end with the smallest letter z . The triangular Lyndon factorization and the shuffle-polynomial theorem therefore give

$$\tilde{\mathcal{A}} = \mathcal{A}[z, x].$$

All these generators are homogeneous in the two counts. Setting z, x to zero gives the unique retraction and preserves that homogeneity. The ideal (z, x) is stable under C because $Cz = x$ and $Cx = z$, and is stable under K . Both operators also preserve $\mathcal{A}$. Their commutation with the retraction follows from this direct-sum polynomial decomposition.

For the explicit formula, deleting one initial x and deleting one terminal z are commuting derivations for shuffle multiplication. Their Taylor substitutions set the corresponding polynomial variables to zero. The u th shuffle power of one letter is $u!$ times its concatenation power, canceling the Taylor factorial. This proves (7.8). Each of its terms before cancellation already has the same two letter counts as w ; after cancellation all remaining words are admissible by the retraction property. $\square$

This retraction is a formal polynomial operation. It assigns no analytic value to divergent endpoint integrals. In the original coordinates its substitution is $z = 0$, $c = b$ in shuffle-polynomial coordinates, not a letter-by-letter replacement inside a concatenation word. It is a different equivariant retraction from the earlier choice $z = -b/2$, $c = b/2$ [17]. The present choice retains the two homogeneous letter counts needed below.

7.3 Eulerian lifts and the oriented projector

For a nonempty word set

$$e(w) = \sum_{k=1}^{|w|} \frac{(-1)^{k-1}}{k} \sum_{\substack{w=w_1 \cdots w_k \\ |w_j| > 0}} w_1 \amalg \cdots \amalg w_k, \quad e(1) = 0. \quad (7.9)$$

This is the first Eulerian logarithm for shuffle multiplication and word deconcatenation. It annihilates products of positive-degree factors and satisfies

$$e^2 = e, \quad e(f) - f \in (\tilde{\mathcal{A}}_{>0})^2 \quad (f \in \tilde{\mathcal{A}}_{>0}). \quad (7.10)$$

For completeness, the product-annihilation property follows by viewing e as the convolution logarithm of the identity character into the commutative shuffle algebra: that logarithm is an infinitesimal character. The second statement is immediate from the $k = 1$ summand of (7.9); product annihilation then gives $e^2 = e$. These are also the standard Eulerian-idempotent properties [15]. Reversal permutes the consecutive cuts and reverses the order of their commuting shuffle factors. Hence e commutes with C , as well as with K , directly from its displayed formula. Every term preserves (n_z, n_x) .

Put $E = R_0 e$ and $U = E(\mathcal{A})$. The retraction R_0 and the Eulerian lift both preserve the two counts, so U is bigraded. On a homogeneous element $u \in U$ of bidegree (p, q) define

$$Q_{\uparrow} u = \begin{cases} u, & p > q, \\ Cu, & p < q, \\ (u + Cu)/2, & p = q. \end{cases} \quad (7.11)$$

Thus a pair of exchanged components is represented on the side with more copies of z . Let $E_{\uparrow}(w) = Q_{\uparrow} E(w)$; it can be computed from the counts in w itself, since E preserves them.

Theorem 7.2 (A non-increasing-depth normal form). *Define $\Pi_{\uparrow}(1) = 1$ and*

$$\boxed{\Pi_{\uparrow}(w) = \sum_{k=1}^{|w|} \frac{1}{k!} \sum_{\substack{w=w_1 \cdots w_k \\ |w_j| > 0}} E_{\uparrow}(w_1) \amalg \cdots \amalg E_{\uparrow}(w_k).} \quad (7.12)$$

Then $\Pi_{\uparrow} : \tilde{\mathcal{A}} \rightarrow \mathcal{A}$ is a graded shuffle-algebra homomorphism. On $\mathcal{A}$ it satisfies

$$\Pi_{\uparrow}^2 = \Pi_{\uparrow}, \quad \Pi_{\uparrow} C = \Pi_{\uparrow}, \quad \Pi_{\uparrow} K = K \Pi_{\uparrow}, \quad (7.13)$$

$$\ker(\Pi_{\uparrow}|_{\mathcal{A}}) = \mathcal{J}, \quad \Pi_{\uparrow}(F_D \mathcal{A}) \subseteq F_D \mathcal{A}. \quad (7.14)$$

Consequently

$$H(f) = H(\Pi_{\uparrow} f), \quad f \in \mathcal{A}, \quad (7.15)$$

and $f - g \in \mathcal{J}$ if and only if $\Pi_{\uparrow} f = \Pi_{\uparrow} g$. In general $C \Pi_{\uparrow} \neq \Pi_{\uparrow}$.

Proof. On $\mathcal{A}$, the map E annihilates positive-degree products and is congruent to the identity modulo such products. It is therefore idempotent and its image U maps isomorphically onto $\mathcal{A}_{>0}/(\mathcal{A}_{>0})^2$. Homogeneous lifts of an indecomposable basis freely generate the polynomial algebra $\mathcal{A}$: the change from its Lyndon generators is triangular in weight, with invertible linear part. Thus

$$\mathcal{A} \simeq \text{Sym}(U) \quad (7.16)$$

as a weight-graded and (n_z, n_x) -bigraded algebra. Both C, K act on U because they commute with R_0, e .

For $p > q$, choose a K -eigenbasis of $U_{n,p,q}$ and the transported basis Cu of $U_{n,q,p}$. On $p = q$, diagonalize the commuting rational involutions C, K . Equation (7.11) identifies each transported pair, fixes the C -positive diagonal generators, and kills the C -negative diagonal generators. It therefore extends uniquely to an algebra retraction $P_\uparrow$ of $\text{Sym}(U)$. Its kernel is the ideal generated by $u - Cu$. This ideal equals $\mathcal{J}$: one inclusion follows by using $u \in \mathcal{A}$ in (7.7); for the other, C acts trivially on the quotient in which these generator relations are imposed. This proves the kernel statement and $P_\uparrow C = P_\uparrow$. The selected eigenspaces also give $P_\uparrow K = K P_\uparrow$.

The formula (7.12) is exactly $P_\uparrow R_0$. Indeed convolution exponentiation of (7.9) gives, for every nonempty word,

$$w = \sum_{k=1}^{|w|} \frac{1}{k!} \sum_{w=w_1 \cdots w_k} e(w_1) \Pi \cdots \Pi e(w_k).$$

Apply R_0 , then $P_\uparrow$. Each $E(w_j)$ belongs to U , even if the block is not admissible. To justify this point, use $\tilde{\mathcal{A}} = \mathcal{A}[z, x]$: in degree at least two, $w - R_0 w$ is a sum of products of positive-degree polynomial factors and is annihilated by e . In degree one the extra generators z, x are killed by $R_0 e$ directly. Thus $E(w) = E(R_0 w) \in U$. The resulting expression is exactly (7.12), proving multiplicativity and idempotence.

Finally each nonzero term of $E(w_j)$ has the original count pair (p_j, q_j) . The orientation either retains p_j copies of z , replaces their number by $q_j > p_j$, or averages two terms with equal counts. It never decreases the number of z 's. Shuffle multiplication adds their counts, so every output word in (7.12) has at least $n_z(w)$ copies of z . This proves the filtration statement word by word.

For admissible f , the kernel description gives $f - \Pi_\uparrow f \in \mathcal{J}$, and (7.6) proves the period identity. To see the last assertion, $\Pi_\uparrow(zy) = zy$ but $C\Pi_\uparrow(zy) = yx \neq zy$. Nevertheless $\Pi_\uparrow(yx) = zy$, as required. $\square$

One immediate all-weight example is

$$\Pi_\uparrow(yx^{n-1}) = z^{n-1}y, \quad H(yx^{n-1}) = 2i\beta(n), \quad n \geq 2.$$

Here the second equality also follows from the single Cayley substitution itself. The theorem deals with the full multiplicative ideal and applies to arbitrary word polynomials, not just this example.

7.4 The exact depth filtration of the full ideal quotient

The filtration on $\mathcal{B}$ is the image of $F_D \mathcal{A}$. It admits a completely explicit polynomial description.

Theorem 7.3 (Filtered polynomial generators). *Choose the K -eigenbases and C -paired bases in the proof of Theorem 7.2. Then $\mathcal{B}$ is freely polynomial on the following generators:*

1. one representative of every C -paired basis vector in $U_{n,p,q} \oplus U_{n,q,p}$, with $p > q$;
2. every C -positive basis vector of $U_{n,p,p}$.

Their minimum depths in this formal quotient are respectively $n - p = n - \max(p, q)$ and $n - p$. The depth of a monomial in these polynomial generators is the sum of their listed depths. Consequently these monomials give a basis adapted to every $F_D \mathcal{B}$.

Proof. Polynomial freeness follows from (7.16) by eliminating one generator from each pair and every negative diagonal generator. The polynomial coordinate changes are bigraded before the quotient. For a surviving quotient monomial, one may independently lift each paired factor on either side of its pair. Its greatest attainable z count is therefore the sum of the larger counts $\max(p, q)$. The representative chosen above attains that count, and thus attains the asserted depth.

Conversely, expand a homogeneous element of $F_D\mathcal{A}$ in the bigraded polynomial basis of $\text{Sym}(U)$. Every source monomial has at least $n - D$ copies of z . A surviving quotient monomial in its image has maximum attainable count no smaller than that source count. Hence it belongs to the span of the basis monomials whose listed depth is at most D . Quotient monomials are linearly independent. A linear combination of lower-depth source elements therefore cannot produce a basis monomial with larger listed depth. This proves the assertion for every filtration step and the claimed minimum within the formal quotient. $\square$

We now give finite coefficient formulas for the number of these polynomial generators. For $n \geq 2$, $p, q \geq 0$, $p + q \leq n$, and $\varepsilon \in \{1, -1\}$ put

$$W_{n,p,q}(\varepsilon) = \frac{1}{n} \sum_{d|\gcd(n,p,q)} \mu(d) \binom{n/d}{p/d} \binom{(n-p)/d}{q/d} (2 + \varepsilon^d)^{(n-p-q)/d}. \quad (7.17)$$

The convention is $\gcd(n, 0, 0) = n$. Then the K -even and K -odd dimensions of $U_{n,p,q}$ are

$$a_{n,p,q}^+ = \frac{W_{n,p,q}(1) + W_{n,p,q}(-1)}{2}, \quad a_{n,p,q}^- = \frac{W_{n,p,q}(1) - W_{n,p,q}(-1)}{2}. \quad (7.18)$$

For the diagonal Cayley traces set

$$T_n(\varepsilon; t) = \frac{(-1)^{n-1}}{n} \left[\sum_{\substack{d|n \\ d \text{ modd}}} \mu(d) \varepsilon^n + \sum_{\substack{d|n \\ d \text{ even}}} \mu(d) (2t^{d/2} + 3)^{n/d} \right], \quad T_{n,p}(\varepsilon) = [t^p] T_n(\varepsilon; t). \quad (7.19)$$

Define

$$c_{n,p}^+ = \frac{T_{n,p}(1) + T_{n,p}(-1)}{2}, \quad c_{n,p}^- = \frac{T_{n,p}(1) - T_{n,p}(-1)}{2}.$$

These are the traces of C on the corresponding K -even and K -odd parts of $U_{n,p,p}$. In particular,

$$c_{n,p}^- = 0 \quad (n \geq 2). \quad (7.20)$$

Indeed for odd $n > 1$ the entire polynomial (7.19) vanishes by $\sum_{d|n} \mu(d) = 0$; for even n it is independent of ε .

Let $g_{n,\delta}^\pm$ count the K -even/odd polynomial generators of weight n and minimum depth δ . For $n \geq 2$ they are

$$g_{n,\delta}^\pm = \sum_{\substack{p>q\geq 0 \\ p+q\leq n \\ n-p=\delta}} a_{n,p,q}^\pm + \sum_{\substack{2p\leq n \\ n-p=\delta}} \frac{a_{n,p,p}^\pm + c_{n,p}^\pm}{2}. \quad (7.21)$$

At weight one the only surviving generators are b and y , so $g_{1,1}^+ = g_{1,1}^- = 1$.

Justification of the character formulas. In weight $n \geq 2$, deleting the degree-one shuffle generators z, x does not change indecomposables. Their character is the free-Lie character on the five-letter space. The classical Witt trace formula [16], or equivalently PBW followed by formal logarithms and Möbius inversion, is

$$\text{tr}(g \mid L_n(V)) = \frac{1}{n} \sum_{d|n} \mu(d) \text{tr}(g^d \mid V)^{n/d}.$$

Apply it first to the diagonal operator with eigenvalues $u, v, \varepsilon, 1, 1$. Extracting $u^p v^q$ gives (7.17); projection to the two K eigenspaces gives (7.18).

On indecomposables, reversal acts by $(-1)^{n-1}$. This follows from the shuffle antipode $S(w) = (-1)^n \text{rev}(w)$ and the fact that the antipode is minus the identity modulo positive-degree products. Consequently C has the character of $(-1)^{n-1} \tau$ there. For the weighted operator combining τ with K if $\varepsilon = -1$, an odd power has trace ε on the one-letter space: the z, x block has zero trace, and the h, b contributions cancel. An even d th power has trace $2(uv)^{d/2} + 3$. Insert these expressions into the Witt trace formula and put $t = uv$. Its coefficients are diagonal traces because a component with $p \neq q$ is carried to a different bidegree. This is exactly (7.19). The same character calculation is valid on the dual indecomposables: transpose has the same trace, including for the weighted operator. Formula (7.21) now counts a full space for $p > q$ and the positive half, with its trace correction, for $p = q$. $\square$

The total Hilbert series and the K -trace series of the associated depth-graded quotient are

$$\mathcal{H}(t, u) = \prod_{n, \delta \geq 1} (1 - t^n u^\delta)^{-g_{n, \delta}^+ - g_{n, \delta}^-}, \quad (7.22)$$

$$\mathcal{H}_K(t, u) = \prod_{n, \delta \geq 1} (1 - t^n u^\delta)^{-g_{n, \delta}^+} (1 + t^n u^\delta)^{-g_{n, \delta}^-}. \quad (7.23)$$

Thus the imaginary, or K -odd, dimensions are the coefficients of $(\mathcal{H} - \mathcal{H}_K)/2$. All products are formal power series; every fixed-weight coefficient is a finite calculation. The first rows, by exact graded depth, are

weight	1	2	3	4	5	6	7	total
1	1							1
2	1	2						3
3	1	9	6					16
4	1	16	40	15				72
5	1	22	116	161	42			342
6	1	28	219	640	593	113		1594
7	1	34	347	1582	3115	2123	316	7518

The total 7518 agrees with the earlier full-ideal count [17]. Its refinement gives

$$\dim F_4 \mathcal{B}_7^- = 1 + 34 + 347 + 1582 = 1964. \quad (7.24)$$

The original imaginary admissible space of weight seven and depth at most four has dimension 2546 [18]; hence

$$\dim(\mathcal{J}_7^- \cap F_4 \mathcal{A}_7) = 582. \quad (7.25)$$

These are dimensions of a formal quotient and of a formal ideal intersection. They do not count linearly independent complex numbers.

7.5 Completeness of the unmultiplied balanced rows

One narrower consequence is useful for interpreting a bounded search. Let $\mathcal{L} = (1 - C)\mathcal{A}$ denote the linear span of unmultiplied Cayley rows, without taking their ideal. For fixed weight n and $k = n - D \geq 0$, let $B_{n, k}$ be the span of admissible adapted words with at least k copies of each of z, x .

Proposition 7.4.

$$\mathcal{L}_n \cap F_D \mathcal{A}_n = (1 - C)B_{n, k}.$$

Thus the balanced seeds exhaust all linear combinations of unmultiplied Cayley rows whose resulting vector has depth at most D , even if higher-depth terms were canceled before taking that result.

Proof. Since C is an involution in characteristic zero, $\mathcal{L} = \ker(1 + C)$. A vector r in the left side satisfies $Cr = -r$, so both r and Cr belong to $F_D\mathcal{A}_n$. The intersection $F_D\mathcal{A}_n \cap C(F_D\mathcal{A}_n)$ has exactly the balanced adapted-word basis $B_{n,k}$. Therefore $r = (1 - C)(r/2)$ is in the right side. The reverse inclusion follows because both a balanced word and its transform have depth at most D . $\square$

At weight seven and depth four the imaginary balanced words have three z 's, three x 's, and one y . There are 50 such admissible words: 140 unrestricted permutations, minus 60 beginning with x , minus 60 ending with z , plus 30 satisfying both exclusions. Exactly four are fixed by C . All fixed signs are positive. Their anti-invariant space therefore has dimension $(50 - 4)/2 = 23$. This completeness statement concerns the unmultiplied linear span. The full ideal intersection has dimension 582 by (7.25); products must be included to obtain that larger object. It does not assert that a previously enumerated collection of bounded products was already complete.

7.6 Consequences for the frozen S_6 search and verification

For any specified family of additional exact relation rows $\mathcal{R} \subset \mathcal{A}$, Theorem 7.2 gives the exact equivalence

$$F \in \mathcal{J} + \text{span}_{\mathbb{Q}} \mathcal{R} \iff \Pi_{\uparrow} F \in \text{span}_{\mathbb{Q}} \{\Pi_{\uparrow} R : R \in \mathcal{R}\}. \quad (7.26)$$

If F and the additional rows have depth at most D , both sides of the projected linear-algebra problem stay in that same depth bound. This is a concrete way to impose the complete Cayley ideal before adding a prescribed double-shuffle or distribution family.

For a reproducible example, freeze the inherited word polynomial [17, Eq. `cay:eq:target`]

$$\begin{aligned} F = & -179712000z^5aa + 485683200z^5a\bar{a} - 36864000z^3az^2a + 401080320zaz^4a \\ & - \frac{47600087040}{61}z^6a + 11750400(za \amalg z^4c) + 109347840(z^3a \amalg z^2c) \\ & - 971366400(z^5a \amalg b). \end{aligned}$$

All powers inside words are concatenation powers. The imaginary part of $H(F)$ is 485683200 times the frozen conjectural S_6 residual. The oriented normal form of $F^- = (F - KF)/2$ has 30 nonzero adapted-word coefficients, all of depth at most two. The previous averaged output has 3444 original-word coordinates. The coordinate systems and representatives differ; these support counts are descriptions of the supplied forms, not minimal-support theorems. The new output is still nonzero. For example,

$$[z^5yh] \Pi_{\uparrow}(F^-) = -166348800, \quad F^- = (F - KF)/2.$$

Its full rational vector is supplied in `s6_oriented_normal_form.json`. This remains an obstruction only for the Cayley ideal, already known without the depth refinement. It neither proves nor disproves the numerical S_6 identity.

The accompanying code uses integers and rational fractions. It compares the consecutive-cut Eulerian formula with its independent permutation formula, verifies the endpoint retraction and all projector identities on all 780 words of weights one through four, and checks 2150 shuffle products of total weight at most four. An independent fraction-free integer elimination builds the entire ideal generated by all Cayley rows and their products through weight four. At every depth its ranks agree with (7.22)–(7.23). These finite audits check implementation and conventions. The proofs above establish the statements in every weight.

A further useful computation is now well specified: apply the oriented projector to the bounded weight-seven double-shuffle and distribution rows, then search for the target in their rational span. The projected problem can use the 1964-dimensional image in (7.24), without adding artificial higher-depth coordinates merely to impose the Cayley ideal. An exact rational certificate is still required before changing the conjecture's status.

8 Source audit, proposed corrections, and integration decisions

The source comparison did not uncover a false theorem in the particular latest Euler, Lerch, and full-Cayley sections used as premises here. It did reveal several places where reading an older fragment in isolation would give an incorrect impression of the project’s current status. The following proposed actions distinguish a newly proved conjecture, an extension of a correct theorem, and an editorial status update.

8.1 Concrete source actions

Source item	Status at the pinned baseline	Proposed action
Near-critical Euler crossing	Explicit conjecture <code>research:conj:crossing</code> in [11].	Replace the conjecture by Theorem 4.1; retain the compact positive b -range and fixed derivative order in its statement.
Lerch sign-block question	The latest report gives a sufficient bound and asks for the least length [12].	Insert Theorem 5.1. The bound is optimal; the reciprocal angle case has length $\pi/\theta + 1$.
Higher critical sums	The first weighted sum is proved; higher sums are proposed in [12].	Insert Theorem 6.1 and the explicit $p = 2$ formula. Retain the conjugate-singularity subtractions.
Full Cayley ideal	A full projector, all-depth dimensions, and the frozen target’s nonmembership are already proved [17].	Add the depth-preserving retraction and filtered counts. Keep $\Pi C = \Pi$ distinct from $C\Pi = \Pi$.
Classical Stieltjes indices 6, 7	Complete all-derivative transition counts are already in the manuscript [13].	Mark any older first-derivative conjectures as superseded. Do not present those counts as a new result of this report.
S_4 , S_6 , and current S_8	S_4 is proved in later material; the current S_6 , S_8 candidates remain conjectural [10].	Preserve these statuses. The 30-term normal form is a formal reduction of the S_6 target, not a period certificate.

For clarity, the already established Stieltjes result is

$$\#\{a > 0 : \gamma_6^{(k)}(a) = 0\} = \begin{cases} 4, & 1 \leq k \leq 3, \\ 6, & k \geq 4, \end{cases} \quad \#\{a > 0 : \gamma_7^{(k)}(a) = 0\} = \begin{cases} 5, & 1 \leq k \leq 4, \\ 7, & k \geq 5. \end{cases}$$

All these zeros are simple, with the global count justified by the source’s analytic bound and rational sign certificates. This report does not replay those inherited certificates and does not claim their discovery. Likewise, the dominant inverse pair for the Lerch velocities is an inherited selected-sheet theorem, not an open assumption.

8.2 Proof obligations that should remain explicit

Joint asymptotics require a joint remainder. The fixed-parameter Euler density expansion is correct within its stated scope. Substituting $a = 1 - \varepsilon$ into a remainder whose constant can depend on a would not prove the conjectured crossing law. The algebraic coefficient tends to zero and the seed survives. The new Lemma 4.3 supplies the missing ε factor; Lemma 4.4 controls both competing scales separately. The proposed correction is to any attempted use beyond the old theorem’s scope, rather than to that theorem itself.

Sharp finite parts see both dominant singularities. The first weighted Lerch sum has no divergent conjugate contribution. At the next order the conjugate term has magnitude $\sqrt{N}$. Consequently a proposed higher-order finite part subtracting only the nonoscillatory powers cannot converge in general. Equation (6.4) is the required addition. This is a concrete correction to the naive extension suggested by the first-order pattern; it is not a claim that the source stated that extension as a theorem.

Depth is attached to a specified relation system. The quotient in Section 7 imposes the full Cayley ideal, and its depth minima are proved within that quotient. Extra shuffle, stuffle, distribution, functional, or period relations can identify additional elements. Neither a formal dimension nor an unsuccessful bounded search gives a lower bound for the dimension of numerical periods. The source ledger already emphasizes this distinction [10]; the proposed integration preserves it.

Translation and endpoint conventions matter. In the Gamma energy the shift is periodic, with a break at $1 - h$. Replacing it by $\log \Gamma(x + h) - \log \Gamma(x)$ over the whole interval changes the problem. For complex spectral parameters the symmetric polylogarithm pair in (2.9) must not be replaced by a real part. The removable pole in $u(u - 1)\zeta(1 + u)$ must be canceled before order differentiation. Finally, repeated definite primitives require the polynomial subtraction in (2.25); an indefinite primitive alone does not fix the endpoint data.

8.3 Suggested placement

The translation and Gamma sections fit the manuscript’s integrated zeta-jet and Stieltjes-antiderivative chapter, with the sharp regularity result cross-referenced from the spectral discussion. The uniform Euler section extends the real-order kernel and fractional continuation. The sign-block and critical-sum results belong immediately after the dominant Lambert-pair theorem. The Cayley section belongs after the full ideal projector, followed by the weight-seven search discussion. Complete source paths and theorem labels, without dependence on changing chapter numbers, are in `integration/claim_ledger.json` and `integration/INTEGRATION.md`.

9 Further research questions and proposed work

The following questions are deliberately stated with a target and a specific obstacle. None is used in the proofs above. The first group extends the new analytic identities; the later groups connect them back to the remaining arithmetic and geometric questions in the source.

9.1 Translation identities and the Stieltjes hierarchy

Question 9.1 (Rational shifts and explicit character coordinates). For $h = p/q$, produce a systematic normal form for the translated pairings in terms of Dirichlet L -function order derivatives and conductor logarithms. Which low-conductor specializations simplify to small explicit Gamma, polygamma, and zeta identities?

The exact finite Fourier identity

$$\mathrm{Li}_s(e^{2\pi ip/q}) = q^{-s} \sum_{a=1}^q e^{2\pi ipa/q} \zeta(s, a/q), \quad \Re s > 1,$$

and its order derivatives already provide a finite conversion. The next step is to combine it with the repository’s conductor and distribution reductions, retaining the logarithmic terms generated

by q^{-s} . This is a coordinate problem; it should not be confused with proving that the resulting constants are independent.

Question 9.2 (Higher translation energies). For $r \geq 2$, determine the extrema and concavity intervals of $V_r(h) = \int_0^1 [\Delta_h \zeta^{(r)}(0, x)]^2 dx$. Does the half-period remain the unique global maximizer in every order?

Symmetry ensures a stationary point at $1/2$, but a positive cosine-series coefficient does not by itself force that point to be the global maximum. The master identity gives the complete derivative formula. A proof would need a suitable sign theorem for combinations of higher Stieltjes constants, extending the elementary $r = 1$ argument.

Question 9.3 (Exact endpoint Gram determinants). Find closed formulas, recurrence relations, or asymptotic bounds for the determinants of the endpoint-polynomial Gram matrices, and for the orthogonal polynomials associated with their two-component integral representation.

Their dependence on the logarithmic shift is triangular and their positivity is exact. The unexplained part is the all-order determinant and orthogonalization structure, whose entries contain zeta values. An explicit recurrence could also improve high-order evaluation of the Stieltjes translation covariances.

Question 9.4 (Endpoint regularity beyond the Sobolev threshold). Determine the complete Besov and logarithmic endpoint regularity of the normalized Stieltjes primitives, including critical indices and finite linear combinations.

The exact law $\|\Delta_h J_n\|_2^2 \sim h \log^{2n+2}(1/h)/(n+1)^2$ already determines the first critical obstruction. Higher finite differences and the explicit Fourier polynomials should determine the remaining scales. A useful outcome would specify convergence rates for Fourier and spline approximations without assuming bounded endpoint derivatives.

9.2 Uniform Euler limits

Question 9.5 (Effective reversal bounds). Replace the asymptotic remainders in Lemma 4.4 by explicit constants on prescribed b intervals. Can the corrected Lambert expression be turned into a certified bracket for ν_r at practical ε values?

The proof already separates the error budgets. The task is to retain numerical constants in the seed-moment, central-window, and tail bounds. An exact positive-integral evaluation can then confirm signs at the two proposed bracket endpoints. A small quadrature residual alone would not certify the bracket.

Question 9.6 (All-order and growing-parameter transitions). Obtain an arbitrary-order expansion of the near-critical reversal and identify the valid joint ranges when r , b , or $1/b$ grows with $\lambda = \log(1/\varepsilon)$.

For fixed r, b , every further density moment and Gaussian logarithmic moment is explicit in zeta and polygamma values. Growing r moves the Gaussian moment's mass, and $b \downarrow 0$ changes the small-variable seed bounds. These effects need a fresh localization. In particular, the fixed- r ratio $\nu_r/\nu_0 \sim [4^r / \binom{2r}{r}]^2$ cannot be used uniformly in growing r without that analysis.

Question 9.7 (Finite-index Euler extremizers). Resolve the latest continuation's all-index axis and kernel-uniqueness conjectures, beyond their already proved initial and eventual cases.

The question is the location of a global maximizer over both orders. It is stronger than the eventual localization theorem, but different from a pointwise comparison at fixed inner order. A practical route is an effective eventual threshold plus exact algebraic or interval analysis for the remaining finite indices. The new near-critical reversal describes one potentially delicate boundary regime.

9.3 Lerch phase arithmetic and critical sums

Question 9.8 (Phase arithmetic and quantitative sign statistics). Is $\theta(2)/(2\pi)$ irrational? More generally, for algebraic $A > 1$, which arithmetic properties of $\theta(A)$ control the frequency and length distribution of same-sign velocity runs?

The optimal block theorem needs no answer to this arithmetic question. Sharper statistics do: equidistribution would give sign frequencies, while quantitative bounds near the cosine zeros require both a Diophantine estimate and the additive, rather than relative, coefficient asymptotic. The reciprocal-angle cases also invite a classification of the corresponding A values.

Question 9.9 (Uniform and smoothed critical finite parts). Develop the critical-sum theorem uniformly as $A \downarrow 1$ or $A \rightarrow \infty$, and compare its sharp-cutoff constant with constants obtained from exponential or smooth cutoffs.

The two singularities move with A , and the factor $(1 - e^{-2i\theta})^{-1}$ becomes large as the phase approaches an endpoint. The fixed- A theorem does not cover this coalescence. Different cutoffs may retain the same local analytic jet but alter subtracted terms; any equality of the resulting constants must be proved from the corresponding generating operator.

Question 9.10 (Global indexing between root regimes). Prove the indexed bulk phase formula proposed in the latest Lerch continuation, with its exact half-integer shift, and match the bulk zeros to the small-root Bessel and large-root Gamma regimes.

The additive cosine expansion locates a local root near a phase value. It does not on its own identify that phase with the globally ordered j th zero. A uniform overlap argument with endpoint asymptotics is needed to fix the index. This is also a route to discrepancy estimates for the existing weak zero distribution.

9.4 Exact polylogarithmic reductions and formalization

Question 9.11 (An exact certificate for the frozen mixed S_6 target). After imposing the complete Cayley ideal with $\Pi_{\uparrow}$, does the 30-term target lie in the span generated by a precisely stated double-shuffle, distribution, and boundary relation system? If so, produce an exact rational certificate.

At weight seven and depth at most four the ambient projected problem has dimension 1964. This makes a finite calculation more focused, but no dimension count ensures that the target vanishes there. If that specified system fails, the output should be an explicit linear functional separating the target from that system, not a claim of numerical nonreducibility. The current S_8 candidate needs the same distinction between exact relation and tiny residual.

Question 9.12 (Relations compatible with the oriented depth basis). Construct effective multiplication and change-of-basis algorithms in the filtered Cayley quotient, and determine which additional exact polylogarithm relations admit similarly compatible depth projections.

The generator theorem gives existence and minimum formal depth. The remaining algorithmic question is complexity at large weights, where the raw Eulerian permutation sum is expensive. Lyndon recursion or consecutive-cut dynamic programming may provide practical substitutes. Any extension should specify its ideal before claiming completeness.

Question 9.13 (Certified evaluation and proof-assistant targets). Formalize the exact finite identities first, and then connect them to the analytic theorems through a small collection of reusable lemmas.

Natural finite targets are the endpoint retraction, shuffle projector, character counts, and polynomial coefficient extraction. Analytic targets include the L^2 holomorphic Fourier formula, the removable Laurent products, the epsilon-sensitive density estimate, and transfer from a finite collection of slit singularities. The supplied scripts are reproducible checks and prospective formalization inputs, rather than completed proof-assistant developments.

10 Reproducibility and verification record

The package includes the modular article sources, an assembled single TeX file, the rendered PDF, exact-algebra programs, high-precision diagnostics, source provenance, and integration notes. Its README gives complete commands and separates the quick exact checks from the slower integral diagnostics. A SHA-256 manifest identifies the delivered files; rebuilding a PDF may change its metadata and therefore its binary hash.

Analytic identities. The translation program compares ordinary split Gamma quadrature against the Hurwitz representation at several shifts, including a small shift. It checks the two-order germ at real and complex orders, the odd-zeta series with its analytic tail bound, the accelerated γ_1 formula, and the second derivative. Rational-shift polylogarithm order derivatives are evaluated independently through a finite Hurwitz Fourier transform. Exact symbolic truncation constructs the endpoint polynomials, rather than fitting their coefficients. The supplement checks the Gram integral and the explicit primitive.

Euler transition. Nine samples evaluate the exact compensated kernel for $b = 1, 2$ and derivative orders $r = 0, 1, 2$. They compare root positions with the leading, corrected, and Lambert approximations. A tighter-tolerance repeat changed one logarithmic root by approximately 4.1×10^{-12} . This is useful numerical evidence of stability; the quadrature is not outward-rounded interval arithmetic. The asymptotic theorem rests on the uniform proof, not these samples.

Lerch sums. The critical-sum program derives Puiseux coefficients symbolically, compares two independent velocity recurrences at high precision, and checks the $p = 2, 3, 4$ subtractions on long finite sums. It also displays the growing oscillation left by omitting the conjugate term. The $O(N^{-1/2})$ theorem comes from the proved singular expansion and continuation, not an extrapolation of the diagnostic errors.

Cayley algebra. All acceptance calculations in the projector suite use integers and rational fractions. It tests 780 words and 2150 products, compares two Eulerian formulas, and independently builds the entire ideal in weights at most four. The independently obtained ranks at every depth agree with the character formulas. The full frozen S_6 vector is included with its rational normal form and exact nonzero coefficient. These finite tests check the implementation; the all-weight conclusions follow from the polynomial-algebra proofs.

Review status. The delicate Euler remainder, root inversion, and global maximum identification received a separate analytic audit. The Cayley kernel and exact minimum-depth argument received an independent algebraic review. The final translation Gram and primitive identities were also checked independently. These reviews are documented in the package. The report was prepared with AI assistance and has not undergone external peer review. Its claims are ordinary mathematical proofs with stated dependencies and reproducible evidence, ready for the project’s integration and further mathematical review.

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